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Master Thesis
Added Resistance of a Ship Navigating in Ice Floes

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Abstract

The prediction of ship resistance in ice-covered waters is a critical challenge for the maritime industry, as Arctic shipping routes become increasingly accessible due to climate change. Model-scale towing tank tests, while valuable, are expensive and of limited repeatability when ice floes are involved. Numerical simulation offers a cheaper and more flexible alternative.

The aim of this thesis is to use the commercial CFD–DEM software Simcenter STAR-CCM+ to calculate the added resistance of a ship advancing through a field of floating ice floes. The fluid flow is governed by the RANS equations and the free surface is captured using the Volume of Fluid (VOF) method. Ice floes are modelled as discrete cylindrical DEM particles, enabling explicit resolution of hull–floe and floe–floe collisions. A large part of the work has been focused on the generation of realistic ice-floe fields, the calibration of contact parameters, and the correction of hydrodynamic forces acting on the floes.

The CFD–DEM setup is validated against published experimental data using the KCS (KRISO Container Ship), over a range of speeds (F_r = 0.06–0.18) and ice concentrations of 60%, 70%, and 80%.

The results show that the model predicts the added ice resistance with relative errors below 10% at high speeds and moderate concentrations. At low speeds and high concentrations, the model underestimates the resistance due to premature floe rotation. Despite these limitations, the model demonstrates that CFD–DEM can be used as a reliable tool to investigate the added resistance of ships navigating in ice floes.

Contents

Nomenclature

B, L_{pp}, T	Ship breadth / length between perp. / draft	m
S	Wetted surface area of ship	m^2
ρ	Fluid density (water, air)	$\mathrm{kg\,m^{-3}}$
μ	Fluid dynamic viscosity (water, air)	$\mathrm{Pa\,s}$
g	Gravitational acceleration	$\mathrm{m\,s^{-2}}$
$\overline{\mathbf{u}}$	Time-averaged fluid velocity	$\mathrm{m\,s^{-1}}$
$\overline{p}$	Time-averaged pressure	Pa
$\overline{\boldsymbol{\tau}}$	Viscous stress tensor	Pa
μ_t	Turbulent (eddy) viscosity	$\mathrm{Pa\,s}$
k, ω	Turbulent kinetic energy / dissipation rate	—
α_w	Water volume fraction	—
y^+	Wall distance	—
F_r, R_e	Froude number / Reynolds number	—
u_τ, τ_w	Friction velocity / wall shear stress	$\mathrm{m\,s^{-1}}$, Pa
T_{prism}, Δy	Prism layer thickness / first cell height	m
R, t_{ice}	Ice floe radius / thickness	m
N, c_{ice}	Number of floes / ice concentration	—
S_{ice}	Surface area of the ice domain	m^2
ρ_{ice}	Ice density	$\mathrm{kg\,m^{-3}}$
E_{ice}, ν_{ice}	Young's modulus / Poisson's ratio of ice	—
μ_{lg}, σ_{lg}	Log-normal distribution parameters	—
A_p, V_p	Projected area / volume of particle	—
m_p, M_{eq}	Particle mass / equivalent mass	kg
$\mathbf{v}_p$, $\mathbf{v}_s$	Particle velocity / slip velocity	$\mathrm{m\,s^{-1}}$
$\boldsymbol{\omega}_p$, $\boldsymbol{\omega}_{\mathrm{rel}}$	Particle / relative angular velocity	$\mathrm{rad\,s^{-1}}$
I_p, I_{eff}, A_θ	Particle / effective / added moment of inertia	$\mathrm{kg\,m^2}$
S_x, S_y	Moment-of-inertia scaling factors	—
C_D, $C_{D,\theta}$, C_{vm}	Drag / rotational drag / virtual mass coeff.	—
$\mathbf{F}_{\mathrm{D}}, \mathbf{F}_{\mathrm{p}}, \mathbf{F}_{\mathrm{vm}}, \mathbf{F}_{\mathrm{g}}$	Drag / pressure / virtual-mass / gravity force	N
$\mathbf{M}_b$	Hydrodynamic drag torque	$\mathrm{N\,m}$
$\mathbf{F}_{\mathrm{c}}$	Contact force	N
F_n, F_t	Normal / tangential contact force component	N
$\mathbf{M}_c$	Contact torque	$\mathrm{N\,m}$
K (K_n, K_t)	Spring stiffness (normal / tangential)	$\mathrm{N\,m^{-1}}$
N_n, N_t	Normal / tangential damping coefficient	$\mathrm{N\,s\,m^{-1}}$
η_n, η_t	Normal / tangential damping ratio	—
C_{rest}, C_{fs}	Restitution / static friction coefficient	—
d_n, d_t	Normal overlap / tangential displacement	m
δ_{ij}, $\delta_{\max}$	Floe overlap / max. admissible overlap	m
R_{ice}, R_{OW}, R_T	Ice / open-water / total resistance	N
$F_{r,P}$	Ice (pack ice) Froude number	—
C_P	Pack ice resistance coefficient	—
C_1, C_2	Empirical fitting coefficients (Guo formula)	—
n	Ice concentration exponent	—
F_{RMS}	RMS of DEM contact force on hull	N
T_{avg}	Averaging period for RMS computation	s
F_{DEM}	DEM force on ship hull	N
$I(t)$	Cumulative DEM impulse	$\mathrm{N\,s}$
t, dt, dt_{DEM}	Time / CFD time step / DEM time step	s
t_{contact}, t_{sim}	Duration of ice-floe contact / total simulation time	s
t_{scale}	DEM time scale factor	—
τ_1	Particle contact time scale	s

1 Introduction

1.1 Research motivation

Climate change is reshaping the prospects for navigation in the regions near the poles. The changing conditions in these parts of the world are driving the various actors of the maritime industry to develop methods for navigating efficiently and safely in these waters.

The opening of these new waterways offers opportunities for transit shipping, for the exploitation of resources such as oil, gas, minerals and fisheries [1], and, in the near future, for routing container ships from China to Europe through the Arctic [2]. This new route for container ships could drastically reduce the transit time — from roughly 28 to 18 days — thereby substantially lowering operating costs, while also helping to avoid the geopolitically unstable regions that disrupt world trade [3].

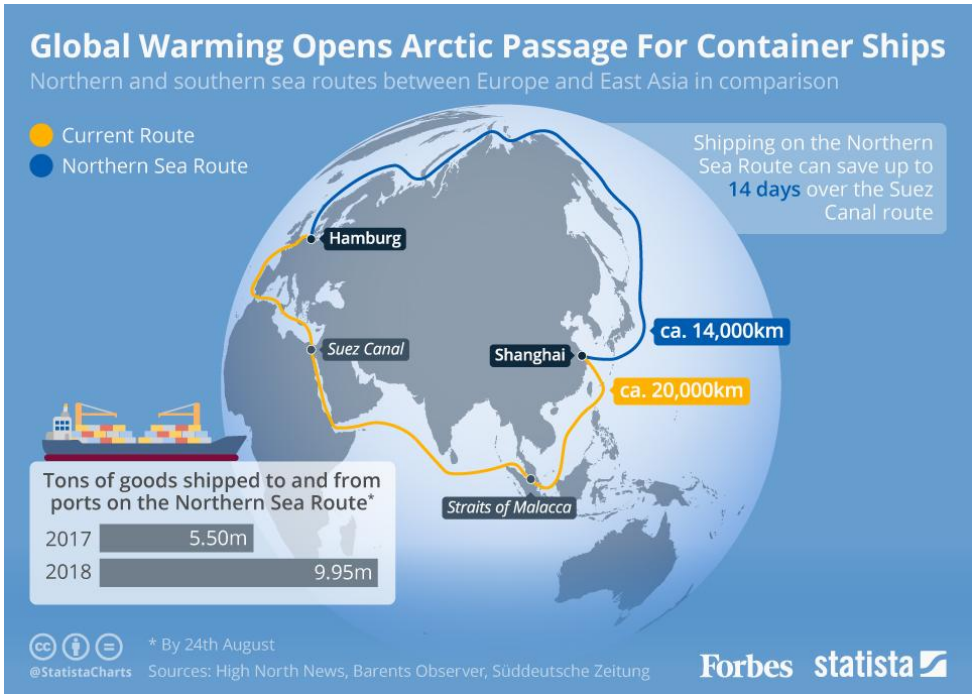


Figure 1: Northern and Southern sea routes between Europe and East Asia in comparison [4]

The efficient exploitation of these waterways relies on understanding the impact of the ice present at the free surface on ships, both structurally and in terms of the resistance to advance, relative to the more classical clean-water case. The free surface can be covered by ice in various forms — pack ice, ice floes or brash-ice channels, for example — and each condition imposes different loads and resistances on the ship. A poor understanding of these loads can lead to premature wear or even to the risk of severe damage (Figure 2).

Figure 2: Bulbous bow damaged by ice during navigation [5]

The case of interest here is that of ice floes, as these are expected to become the form of ice most frequently encountered by ships in the future, for both Arctic and Antarctic shipping [6]. Numerous studies, both experimental and numerical, have been published in recent years on this topic [7, 8, 9, 10], underlining the importance of the ice-added resistance acting on the ship. Such experimental campaigns are, however, extremely expensive, and are therefore not the most practical means of providing easy, low-cost predictions of the resistance of a ship navigating in ice floes.

A solution to this problem can be found in numerical simulation. Indeed, whereas setting up a towing-tank test can cost several thousand euros, a CFD simulation can prove considerably less expensive, which motivates the numerical approach adopted in this work.

Figure 3: Example of a ship navigating in ice floes [11]

1.2 Studies of ship–ice interaction

The interaction between a ship and broken ice has been investigated both experimentally and numerically.

Model-scale testing in ice tanks has long been the primary tool for assessing ship performance in ice. Because natural sea ice is difficult to scale correctly, broken-ice conditions are usually reproduced in towing tanks with synthetic ice. Guo et al. [7] tested a KCS model advancing through a synthetic pack-ice channel and measured the total resistance together with the heave and pitch motions over a range of speeds and ice concentrations, observing that the ratio of the pack-ice to the open-water resistance increases with concentration and decreases with speed. Luo et al. [8] examined the resistance and the motion-attitude variations produced by the combined ship–wave–ice interaction in the marginal ice zone, while Kim et al. [9] compared the resistance of an icebreaking cargo vessel for different waterline angles, showing that the bow geometry strongly affects the ice resistance. These experiments characterise how the ice floes increase the resistance and alter the motions of the hull, but they are costly, time-consuming and, in broken ice, of limited repeatability, since the random arrangement of the floes changes continuously during a run.

Numerical simulation offers a cheaper and more detailed alternative. Finite-element approaches based on explicit dynamics have been used to compute the ice loads, for example with LS-DYNA for an icebreaking cargo vessel in pack ice [12] and with a coupled Eulerian–Lagrangian formulation for ice impacts on hulls in broken-ice fields [10]. Guo et al. [13] also studied the KCS model through numerical simulation using the LS-DYNA software under the same conditions as the experimental study.

More recently, the coupling of Computational Fluid Dynamics (CFD) with the Discrete Element Method (DEM) has become the leading approach for broken-ice resistance, as it resolves both the free-surface flow and the discrete ship–ice and ice–ice collisions. Vroegrijk [14] first validated such a model in STAR-CCM+ against measured data. Huang et al. [15] studied wave interaction with a single circular floe and subsequently developed a CFD–DEM model that incorporated the ship-generated waves and analysed the dependence of the ice-floe resistance on ship speed, ice concentration, ice thickness and floe diameter for the KCS [16]. The same approach has been applied to an ice-strengthened bulk carrier in a brash-ice channel [17] and to full-scale ship–ice interaction under FSICR conditions [18], in good agreement with ice-tank tests.

1.3 Sea-ice characterisation and ice-ice interaction

Beyond the ship itself, the ice has been studied independently, since a realistic description of the floe field is essential for any resistance prediction. Two aspects are particularly relevant here: the geometry and properties of the individual floes, and the way the floes interact with one another.

The floes in a broken field are not uniform in size. Tuovinen [19] showed that the size of the ice blocks in a broken channel follows a log-normal distribution, which is now commonly used to generate representative ice fields in numerical models. The mechanical properties of the ice matter as well: Planque et al. [20] measured the Young's modulus of ice and expressed it as a function of its density, providing input values for the contact models used in the simulations.

Because broken sea ice consists of many discrete bodies that collide, rotate and pack together, its behavior is well described by the Discrete Element Method (DEM), originally introduced by Cundall and Strack for granular assemblies [21]. Hopkins and Tuhkuri [22] used a DEM model to study the compression and ridging of floating ice fields, capturing the contact forces between individual floes, and Damsgaard et al. [23] showed that a particle-based DEM reproduces

the large-scale dynamics of sea ice more faithfully than continuum rheologies. These studies confirm that a discrete description is the natural way to represent the ice floes and their mutual interactions, and they underpin the modelling choices adopted in the present work.

1.4 Objectives and scope

The resistance applied to a ship navigating through ice comprises two components: the hydrodynamic resistance, and the resistance induced by contact with the ice floes. Capturing both aspects simultaneously motivates the coupling of a Computational Fluid Dynamics (CFD) solver with a Discrete Element Method (DEM) solver, which handles the fluid and the discrete ice–hull interactions respectively. The commercial software Simcenter STAR-CCM+ developed by Siemens provides an integrated CFD–DEM framework and is adopted in the present work.

The objective of this thesis is therefore to develop a numerical methodology, based on this CFD–DEM coupling, to investigate the added resistance of a ship advancing through an ice-floe field. The fluid phases are modelled with the RANS equations closed by the k–ω SST turbulence model [24], and the air–water interface is captured with the VOF method [25]. The ice floes are represented as discrete cylindrical Lagrangian particles, allowing hull–floe and floe–floe collisions to be resolved explicitly. The implementation is carried out on the KCS hull, which is widely used as a reference for numerical resistance studies [26], so that the computed ice resistance can be validated against the experimental data of Guo et al. [7]. Particular attention is paid to the generation of realistic ice-floe fields with prescribed concentrations, to the calibration of the DEM contact parameters, and to the known limitations of the CFD–DEM coupling regarding the hydrodynamic surface forces acting on the floes, which are addressed through corrections to the moment of inertia and the drag torque [27, 28]. The floe size distribution and material properties are selected following published data [19, 20].

2 Numerical Methods

This chapter describes the numerical model and its setup.

2.1 Fluid Solver

2.1.1 Navier–Stokes Equations

The fluid motion is governed by the Navier–Stokes equations. In this work, the Reynolds-averaged Navier–Stokes (RANS) equations are solved for an incompressible Newtonian fluid:

$$\nabla \cdot \overline{\mathbf{u}} = 0 \tag{1}$$

$$\frac{\partial(\rho_w \overline{\mathbf{u}})}{\partial t} + \nabla \cdot (\rho_w \overline{\mathbf{u}}\,\overline{\mathbf{u}}) = -\nabla \overline{p} + \nabla \cdot (\overline{\boldsymbol{\tau}} - \mathbf{R}) + \rho_w \mathbf{g} \tag{2}$$

where $\mathbf{u} = \overline{\mathbf{u}} + \mathbf{u}'$ is the instantaneous fluid velocity, $\overline{\mathbf{u}}$ its time-averaged component, $\mathbf{u}'$ the fluctuation, ρ the density, $\overline{p}$ the time-averaged pressure, $\overline{\boldsymbol{\tau}} = \mu\left[\nabla\overline{\mathbf{u}} + (\nabla\overline{\mathbf{u}})^T\right]$ the time-averaged viscous stress tensor, μ the dynamic viscosity, $\mathbf{g}$ the gravitational acceleration, and $\mathbf{R} = \rho\,\overline{\mathbf{u}'\mathbf{u}'}$ the Reynolds stress tensor [29].

2.1.2 Turbulence Modelling

Ship motions and wave–structure interactions can induce turbulent flow, so turbulence effects must be accounted for in the Navier–Stokes equations to obtain accurate results. Averaging the nonlinear convective term in the RANS equations introduces the additional Reynolds stress tensor $\mathbf{R}$, which increases the number of unknowns. A turbulence model is therefore required to close the system. The two most widely used turbulence models are the k–ε and k–ω formulations, each suited to different flow regimes [24]. In the present work, the k–ω SST model is selected, as it combines the robustness of the k–ε model in the far field with the accuracy of the k–ω model in the near-wall region. The principle is presented below using the k–ω model.

Using the Boussinesq approximation, the Reynolds stress tensor is modelled as

$$\mathbf{R} = -\mu_t\left[\nabla\overline{\mathbf{u}} + (\nabla\overline{\mathbf{u}})^T\right] + \tfrac{2}{3}\,\rho\,k\,\mathbf{I} \tag{3}$$

where μ_t is the eddy viscosity and k the turbulent kinetic energy, defined as

$$k = \tfrac{1}{2}\,\overline{u_i' u_i'} = \tfrac{1}{2}\left(\overline{u'^2} + \overline{v'^2} + \overline{w'^2}\right). \tag{4}$$

In the k–ω model, the eddy viscosity is expressed as a function of k and the specific dissipation rate ω:

$$\mu_t = \rho\,\frac{k}{\omega} \tag{5}$$

The variables k and ω are obtained by solving their respective transport equations:

$$\frac{\partial(\rho k)}{\partial t} + \nabla \cdot (\rho\overline{\mathbf{u}}\,k) = \nabla \cdot \left[\left(\mu + \frac{\mu_t}{\sigma_k}\right)\nabla k\right] + P_k - \beta^*\,\rho\,\omega\,k \tag{6}$$

$$\frac{\partial(\rho\omega)}{\partial t} + \nabla \cdot (\rho\overline{\mathbf{u}}\,\omega) = \nabla \cdot \left[\left(\mu + \frac{\mu_t}{\sigma_\omega}\right)\nabla\omega\right] + \gamma\,\frac{\omega}{k}\,P_k - \beta\,\rho\,\omega^2 \tag{7}$$

where P_k is the production term of turbulent kinetic energy, β^*, β, γ, σ_k and σ_ω are model constants [24].

2.1.3 Free-Surface Modelling

Free-surface problems involve two immiscible phases — water and air — and require a method to capture the interface between them. The Volume of Fluid (VOF) method is used here [25].

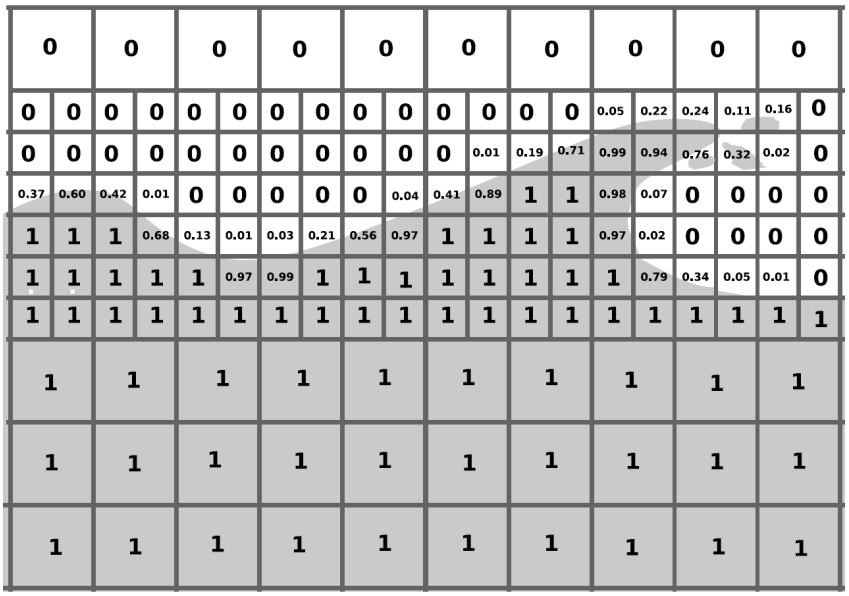


Figure 4: Illustration of free-surface reconstruction in the VOF method [30].

The VOF implementation in STAR-CCM+ predicts the distribution and motion of the interface between immiscible phases. The approach assumes that the mesh resolution is sufficient to resolve the position and shape of the interface. Phase distribution is described by the volume fraction α_w of water in each cell:

$$\alpha_w = \frac{V_{\text{water}}}{V} \tag{8}$$

where V_{water} is the volume of water in the cell and V is the total cell volume [31]. The value of α_w indicates the content of the cell:

- $\alpha_w = 0$: the cell contains only air;
- $\alpha_w = 1$: the cell contains only water;
- $\alpha_w \in (0, 1)$: the cell contains both phases (interface cell).

In the two-fluid VOF formulation, a single velocity and pressure field are solved across the domain. The material properties in interface cells are volume-averaged between the two phases:

$$\rho^{\text{cell}} = \alpha_w \, \rho_w + (1 - \alpha_w) \, \rho_a \tag{9}$$

$$\mu^{\text{cell}} = \alpha_w \, \mu_w + (1 - \alpha_w) \, \mu_a \tag{10}$$

The volume fraction is advected through the transport equation:

$$\frac{\partial \alpha_w}{\partial t} + \nabla \cdot (\overline{\mathbf{u}} \, \alpha_w) + \nabla \cdot [(\overline{\mathbf{u}}_w - \overline{\mathbf{u}}_a) \, \alpha_w (1 - \alpha_w)] = 0 \tag{11}$$

In this study, $\rho_w = 997.561\,\mathrm{kg/m^3}$, $\mu_w = 8.8871 \times 10^{-4}\,\mathrm{Pa\,s}$, $\rho_a = 1.18415\,\mathrm{kg/m^3}$, $\mu_a = 1.85508 \times 10^{-5}\,\mathrm{Pa\,s}$, and $g = 9.81\,\mathrm{m/s^2}$.

2.2 Ice Floe Modelling

2.2.1 CFD–DEM Coupling

STAR-CCM+ provides the capability to couple the CFD solver with a DEM solver, allowing Lagrangian particles to be tracked within an Eulerian fluid flow. In the present work, the ice floes are modelled as DEM particles within the Eulerian fluid domain.

Two types of CFD–DEM coupling are available:

- **One-way coupling**: the DEM particles are influenced by the fluid flow, but they do not affect it in return.
- **Two-way coupling**: the DEM particles also exert forces on the fluid, modifying the flow field.

One-way coupling is selected here, as it provides sufficient accuracy at a substantially lower computational cost. Two-way coupling produces negligible differences in the predicted ship resistance while being significantly more expensive [17].

2.2.2 Lagrangian Description and DEM Formulation

In this simulation, the ice floes are modelled using a Lagrangian approach, in contrast to the surrounding fluids, which are described in an Eulerian framework. In an Eulerian description, flow variables are observed at fixed points in space, whereas the Lagrangian description follows individual entities along their trajectories [29]. This distinction is particularly relevant for CFD–DEM simulations, where the fluid phase is solved on a fixed computational grid while the discrete solid bodies are tracked individually.

A Lagrangian description is necessary for the ice floes because broken sea ice (such as ice floes) behaves as a collection of discrete bodies rather than as a continuous phase. Each floe undergoes its own translation, rotation, and contact interactions with neighbouring floes and with the ship hull. A particle-based Lagrangian approach is therefore better suited to sea ice modelling than continuum rheologies, since individual contact mechanics are more firmly established [23].

The method used to simulate each ice floe is the Discrete Element Method (DEM). The DEM is a numerical method designed to simulate the motion of many interacting discrete solid bodies. It extends the Lagrangian formulation to include inter-particle contact forces in the equations of motion [21]. STAR-CCM+ allows the user to define custom Lagrangian phases and phase-interaction models [32].

2.2.3 Governing Equations for the DEM Particles

The momentum equation for a DEM particle of mass m_p is:

$$m_p \frac{d\mathbf{v}_p}{dt} = \mathbf{F}_\mathrm{D} + \mathbf{F}_\mathrm{p} + \mathbf{F}_\mathrm{vm} + \mathbf{F}_\mathrm{g} + \mathbf{F}_\mathrm{c} \tag{12}$$

where the right-hand-side terms are:

- $\mathbf{F}_\mathrm{D}$, the drag force:

$$\mathbf{F}_\mathrm{D} = \tfrac{1}{2}\, C_D\, \rho_w A_p\, |\mathbf{v}_s| \mathbf{v}_s \tag{13}$$

  where C_D is the translational drag coefficient, A_p is the projected particle area, and $\mathbf{v}_s = \overline{\mathbf{u}} - \mathbf{v}_p$ is the slip velocity between the particle and the continuous phase.

- $\mathbf{F}_{\mathrm{p}}$, the pressure-gradient force:
$$\mathbf{F}_{\mathrm{p}} = -V_p \nabla \overline{p} \tag{14}$$
where V_p is the particle volume.

- $\mathbf{F}_{\mathrm{vm}}$, the virtual mass force:
$$\mathbf{F}_{\mathrm{vm}} = C_{vm}\, \rho_w V_p \left(\frac{D\overline{\mathbf{u}}}{Dt} - \frac{d\mathbf{v}_p}{dt} \right) \tag{15}$$
where C_{vm} is the virtual mass coefficient and D/Dt is the material derivative.

- $\mathbf{F}_{\mathrm{g}}$, the gravitational force.

- $\mathbf{F}_{\mathrm{c}}$, the contact force, defined below.

The angular-momentum equation for the particle is:

$$I_p \frac{d\boldsymbol{\omega}_p}{dt} = \mathbf{M}_b + \mathbf{M}_c \tag{16}$$

with:

- I_p, the particle moment of inertia;

- $\boldsymbol{\omega}_p$, the particle angular velocity;

- $\mathbf{M}_b$, the hydrodynamic drag torque:
$$\mathbf{M}_b = \tfrac{1}{2}\, \rho_w\, R^5\, C_{D,\theta}\, |\boldsymbol{\omega}_{\mathrm{rel}}|\, \boldsymbol{\omega}_{\mathrm{rel}} \tag{17}$$
where R is the radius of the ice floe, $C_{D,\theta}$ is the rotational drag coefficient and $\boldsymbol{\omega}_{\mathrm{rel}} = \frac{1}{2}\nabla \times \overline{\mathbf{u}} - \boldsymbol{\omega}_p$ is the relative angular velocity between the particle and the fluid;

- $\mathbf{M}_c$, the total moment from contact forces.

2.2.3.1 Contact force model. In the DEM model, the contact force $\mathbf{F}_{\mathrm{c}}$ represents both particle–particle (ice–ice) and particle–boundary (ice–ship) interactions. At the contact point, the interaction is represented by a spring–dashpot oscillator.

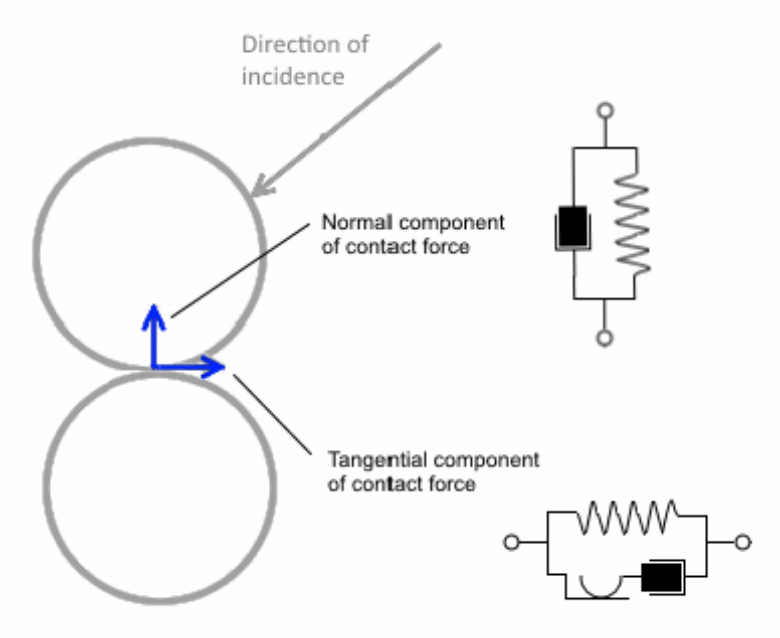


Figure 5: Spring–dashpot oscillator used to model the contact force between two particles. [33]

For the cylindrical ice floes considered here, only the Linear Spring contact model is available in STAR-CCM+ (the other models are made for other shapes, mainly sphere). The contact force between two cylinders A and B is decomposed into normal and tangential components:

$$\mathbf{F}_c = F_n\,\mathbf{n} + F_t\,\mathbf{t} \tag{18}$$

where $\mathbf{n}$ is the contact-point normal unit vector and $\mathbf{t}$ the tangential unit vector in the contact plane.

In the Linear Spring contact model, both for normal and tangential terms, the spring term accounts for the elastic repulsion generated by the overlap between the two particles, while the dashpot term provides viscous damping and dissipates part of the collision energy. The normal and tangential force components are [21]:

- Normal force:
$$F_n = -K_n\,d_n - N_n\,v_n \tag{19}$$
- Tangential force: $F_t = -K_t\,d_t - N_t\,v_t$ as long as $|K_t d_t| < |K_n d_n|\,C_{fs}$, where C_{fs} is the static-friction coefficient; otherwise:
$$F_t = \frac{|K_n d_n|\,C_{fs}\,d_t}{|d_t|} \tag{20}$$

Where d_n and d_t are overlaps in normal and tangential directions at the contact point, v_n and v_t are the normal and tangential velocity components of the relative particle surface velocity at the contact point.

The parameters K_n, K_t, N_n and N_t are defined as follows:

- K_n is the normal spring stiffness constant;
- K_t is the tangential spring stiffness constant;

- $N_n = 2\,\eta_n\,\sqrt{K_n\,M_{\text{eq}}}$, where η_n is the dimensionless normal damping ratio, obtained from the restitution coefficient C_{rest} via (21), and M_{eq} is the equivalent particle mass (22);
- $N_t = 2\,\eta_t\,\sqrt{K_n\,M_{\text{eq}}}$, where η_t is the dimensionless tangential damping ratio, obtained from the restitution coefficient C_{rest} via (21).

$$\eta = \frac{-\ln(C_{\text{rest}})}{\sqrt{\pi^2 + \ln^2(C_{\text{rest}})}} \tag{21}$$

$$M_{\text{eq}} = \frac{1}{\dfrac{1}{M_A} + \dfrac{1}{M_B}} \tag{22}$$

Two key parameters in the DEM contact model are the spring stiffness constants K_n and K_t, which convert the inter-particle overlap into a contact force. Their values are chosen based on a sensitivity test performed in Section 3.6.

The static friction coefficient C_{fs} and the restitution coefficient C_{rest} are purely physical parameters [28].

2.2.4 DEM solver parameters

To obtain the DEM particle trajectory, (12) is integrated over time using a characteristic DEM particle time scale. The particle time step depends on a contact time scale τ_1 and a user-defined scaling factor t_{scale}:

$$\tau_1 = 2\sqrt{\frac{m_p}{K}} \tag{23}$$

where m_p is the particle mass and K is the spring stiffness constant. In the present study, the normal and tangential spring stiffness constants are set equal, $K_n = K_t = K$, so that a single stiffness parameter governs both the contact model and the DEM time step. The DEM time step is then:

$$dt_{\text{DEM}} = t_{\text{scale}}\,\tau_1 \tag{24}$$

The spring stiffness constant K must therefore be selected carefully, as it governs both the physical accuracy and the computational cost of the simulation. A value that is too low results in excessive particle overlap, leading to artificially soft contact behavior and unreliable force predictions. Conversely, an excessively large value imposes a small DEM time step according to (23) and increases the computational cost. In practice, K should be chosen as the minimum value and t_{scale} as the maximum value for which the simulation results become independent of further stiffness increases. A sensitivity study was therefore conducted in this work to identify this convergence threshold, as discussed in Section 3.6.

2.2.5 Limitations of the CFD–DEM Model

The CFD–DEM coupling introduces modelling assumptions that can lead to non-physical behavior for individual ice floes. Two limitations are discussed below: the inability to enforce the correct equilibrium draft, and the simplified representation of hydrodynamic surface forces.

As expressed in (12) and (16), the hydrodynamic forces acting on a DEM particle are computed using simplified force models rather than by integrating the fluid stress over the actual particle surface. Since DEM particles are treated as rigid bodies subject to resultant forces and

torques at their centre of mass, spatially distributed surface loads — such as hydrostatic pressure gradients — cannot be represented in a physically consistent manner.

Regarding the equilibrium draft, this limitation is inherent to the DEM framework: the correct submergence depth of the floe, governed by the density ratio ρ_{ice}/ρ_w, cannot be enforced through the DEM force model alone. No straightforward correction is available within the present coupling strategy, and this represents an acknowledged source of modelling error.

2.2.5.1 Hydrodynamic Torque and Added Moment of Inertia. The hydrodynamic torque associated with added mass is not directly computed by the DEM model. However, its effect can be introduced implicitly by augmenting the particle moment of inertia. STAR-CCM+ provides a scaling factor for the moment of inertia of each particle about its principal axes, which can be exploited for this purpose. Since the added-mass torque is proportional to the angular acceleration of the particle, it is formally equivalent to an increase in rotational inertia. For a rotational degree of freedom, the added-mass torque is:

$$\mathbf{M}_a = -A_\theta \frac{d\boldsymbol{\omega}_p}{dt} \tag{25}$$

where A_θ is the added moment of inertia associated with rotational motion. This contribution is incorporated into the equation of motion by replacing the physical moment of inertia I_p with an effective value:

$$I_{\text{eff}} = I_p + A_\theta \tag{26}$$

The ice floe is assumed to be initially horizontal, with its symmetry axis vertical. Upon contact with the ship hull, it undergoes roll or pitch rotation about a horizontal axis. The corresponding added moments of inertia are equal by axial symmetry and given by [27]:

$$A_{\theta x} = A_{\theta y} = \frac{8}{45}\,\rho_w\,R^5 \tag{27}$$

The physical moments of inertia of a thin cylindrical floe about the horizontal axes are:

$$I_x = I_y = \frac{1}{12}\,m_p\big(3R^2 + t_{\text{ice}}^2\big) \tag{28}$$

where t_{ice} is the floe thickness and $m_p = \rho_{\text{ice}}\,\pi\,R^2\,t_{\text{ice}}$ its mass. In the thin-floe limit $t_{\text{ice}} \ll R$, this simplifies to:

$$I_x = I_y \simeq \frac{1}{4}\,\rho_{\text{ice}}\,\pi\,t_{\text{ice}} R^4 \tag{29}$$

The ratio of added to physical moment of inertia is therefore:

$$\frac{A_{\theta x}}{I_x} = \frac{A_{\theta y}}{I_y} = \frac{32}{45\pi}\,\frac{\rho_w}{\rho_{\text{ice}}}\,\frac{R}{t_{\text{ice}}} \tag{30}$$

Accordingly, the moment of inertia scaling factor to be applied in STAR-CCM+ is:

$$S_x = S_y = \frac{I_p + A_\theta}{I_p} = 1 + \frac{32}{45\pi}\,\frac{\rho_w}{\rho_{\text{ice}}}\,\frac{R}{t_{\text{ice}}} \tag{31}$$

For typical ice and water densities ($\rho_{\text{ice}} \approx 900\,\text{kg/m}^3$, $\rho_w \approx 1025\,\text{kg/m}^3$) and the mean value of the floe geometry used in this study ($R = 0.1\,\text{m}$, $t_{\text{ice}} = 0.02\,\text{m}$), this gives $S_x = S_y \approx 2.25$.

2.2.5.2 Particle Drag Torque. The rotational drag coefficient $C_{D,\theta}$ used in the DEM torque model (17) must be determined for the cylindrical ice floes considered here. It is derived from Morison's quadratic drag equation applied to a thin disk in roll or pitch.

The elementary drag force acting on a surface element dA is:

$$dF_D = \tfrac{1}{2}\,\rho_w\,C_D\,|V_r|\,V_r\,dA \tag{32}$$

For a rigid body rotating with angular velocity $\dot{\theta}$ about the y-axis, a point at horizontal distance x from the rotation axis has a normal velocity:

$$V_r = x\,\dot{\theta} \tag{33}$$

Substituting into the Morison equation, the elementary drag force becomes:

$$dF_D = \tfrac{1}{2}\,\rho_w\,C_D\,\dot{\theta}\left|\dot{\theta}\right|\,x^2\,dA \tag{34}$$

The corresponding elementary drag torque, with moment arm x, is:

$$dM_D = x\,dF_D = \tfrac{1}{2}\,\rho_w\,C_D\,\dot{\theta}\left|\dot{\theta}\right|\,x^3\,dA \tag{35}$$

Integrating over the wetted half-disk $\mathcal{D}^+$ yields the total drag torque:

$$M_D = -\tfrac{1}{2}\,\rho_w\,C_D\,\dot{\theta}\left|\dot{\theta}\right|\int_{\mathcal{D}^+} x^3\,dA \tag{36}$$

where the negative sign reflects the resisting nature of the drag torque, acting in the direction opposite to $\dot{\theta}$.

The half-disk $\mathcal{D}^+$ is decomposed into vertical strips parallel to the y-axis, as illustrated in Figure 6. Since the disk boundary satisfies $x^2 + y^2 \leq R^2$, the strip at a given x extends from:

$$-\sqrt{R^2 - x^2} \leq y \leq +\sqrt{R^2 - x^2} \tag{37}$$

so that the strip height and elementary area are:

$$h(x) = 2\sqrt{R^2 - x^2}, \qquad dA = h(x)\,dx = 2\sqrt{R^2 - x^2}\,dx \tag{38}$$

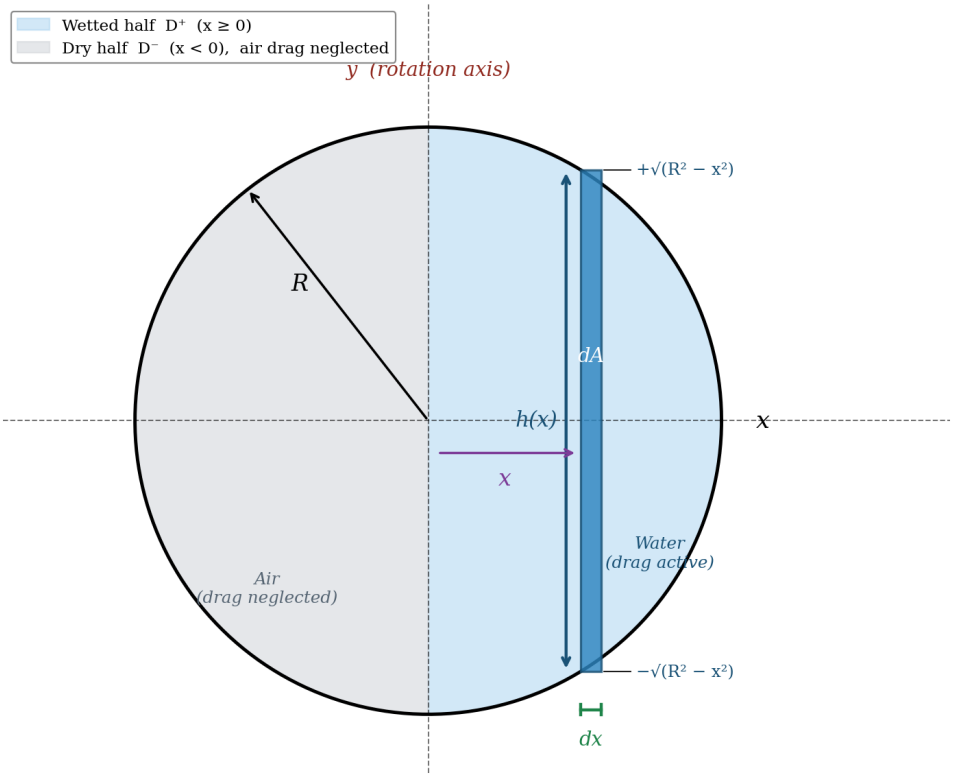


Figure 6: Vertical-strip decomposition of the wetted half-disk $\mathcal{D}^+$ used to evaluate the geometric integral of the drag torque.

The geometric integral in (36) becomes:

$$\mathcal{I} = \int_{\mathcal{D}^+} x^3 \, dA = 2 \int_0^R x^3 \sqrt{R^2 - x^2} \, dx \tag{39}$$

Applying the trigonometric substitution $x = R\sin\varphi$, $dx = R\cos\varphi \, d\varphi$, $\sqrt{R^2 - x^2} = R\cos\varphi$, with bounds $x = 0 \Rightarrow \varphi = 0$ and $x = R \Rightarrow \varphi = \pi/2$:

$$\mathcal{I} = 2 \int_0^{\pi/2} \underbrace{R^3 \sin^3\varphi}_{x^3} \cdot \underbrace{R\cos\varphi}_{\sqrt{R^2-x^2}} \cdot \underbrace{R\cos\varphi \, d\varphi}_{dx} = 2R^5 \int_0^{\pi/2} \sin^3\varphi \cos^2\varphi \, d\varphi \tag{40}$$

Using the identity $\sin^3\varphi = \sin\varphi\,(1 - \cos^2\varphi)$ and the substitution $u = \cos\varphi$:

$$\int_0^{\pi/2} \sin^3\varphi \cos^2\varphi \, d\varphi = \int_0^1 u^2 \, du - \int_0^1 u^4 \, du = \tfrac{1}{3} - \tfrac{1}{5} = \tfrac{2}{15} \tag{41}$$

The geometric integral therefore evaluates to:

$$\boxed{\mathcal{I} = \tfrac{4}{15} R^5} \tag{42}$$

Substituting back into (36), the drag torque on the ice floe at first contact is:

$$M_D = -\tfrac{1}{2} \rho_w \tfrac{4}{15} C_D R^5 \dot{\theta} \left|\dot{\theta}\right| \tag{43}$$

This expression is now compared with the STAR-CCM+ rotational drag torque introduced in (17):

$$\mathbf{M}_b = \tfrac{1}{2}\,\rho_w\,R^5\,C_{D,\theta}\,|\boldsymbol{\omega}_{\text{rel}}|\,\boldsymbol{\omega}_{\text{rel}}$$

By identification of the two expressions, the rotational drag coefficient to be entered in STAR-CCM+ for the cylindrical ice floes considered here is:

$$\boxed{C_{D,\theta} = \tfrac{4}{15}\,C_D} \tag{44}$$

2.3 Summary of Numerical Parameters

This subsection summarises the numerical and physical parameters used in the CFD–DEM simulations of the present study. The values are gathered here for quick reference and reproducibility.

2.3.1 Fluid Properties

The two fluid phases (water and air) are modelled as incompressible Newtonian fluids with the properties listed in Table 1.

Property	Symbol	Value	Unit
Water density	ρ_w	997.561	$\mathrm{kg/m^3}$
Water dynamic viscosity	μ_w	8.8871×10^{-4}	Pa s
Air density	ρ_a	1.18415	$\mathrm{kg/m^3}$
Air dynamic viscosity	μ_a	1.85508×10^{-5}	Pa s
Gravitational acceleration	g	9.81	$\mathrm{m/s^2}$

Table 1: Fluid properties used in the simulations.

2.3.2 Ice Floe Geometry and Properties

The ice floes are modelled as thin cylindrical disks. Their geometric and material properties are listed in Table 2.

Property	Symbol	Value	Unit
Ice density	ρ_{ice}	900	$\mathrm{kg/m^3}$
Poisson's ratio	ν	0.33	—
Young's modulus	E_{ice}	9.0	GPa

Table 2: Ice floe geometry and physical properties.

2.3.3 DEM Hydrodynamic Coefficients

The hydrodynamic force and torque coefficients applied to the DEM particles are summarised in Table 3. The rotational drag coefficient $C_{D,\theta}$ corresponds to the value derived in (44) from the half-disk model.

Coefficient	Symbol	Value	Comment
Translational drag coefficient	C_D	1.2	From [28]
Virtual mass coefficient	C_{vm}	1.0	From [28]
Rotational drag coefficient	$C_{D,\theta}$	$\frac{4}{15} C_D = 0.32$	From (44)
Moment-of-inertia scaling	$S_x = S_y$	≈ 2.25	From (31)

Table 3: Hydrodynamic coefficients used in the DEM model.

2.3.4 DEM Contact Model Parameters

The parameters of the Linear Spring contact model are listed in Table 4. The normal and tangential spring stiffness constants are chosen from the sensitivity test (Section 3.6).

Parameter	Symbol	Value	Unit
Normal spring stiffness	K_n	60000	N/m
Tangential spring stiffness	K_t	60000	N/m
Restitution coefficient	C_{rest}	0.8098	—
Static friction coefficient (ice-ice)	C_{fs}	0.35	—
Static friction coefficient (ice-wall)	C_{fs}	0.035	—

Table 4: DEM contact model parameters.

2.3.5 Numerical Setup

The simulation control parameters (time step, turbulence model, solver schemes) are summarised in Table 5.

Setting	Value
Solver type	Segregated, implicit unsteady
Turbulence model	k–ω SST
Free-surface model	Volume of Fluid (VOF), High-Resolution Interface Capturing
CFD–DEM coupling	One-way
CFD Time step	$dt = 0.01\,\text{s}$
Duration of ice-floe contact	$t_{\text{contact}} = 40\,\text{s}$
Total simulation time	$t_{\text{sim}} = 80\,\text{s}$
Inner iterations / step	5

Table 5: Numerical setup of the CFD–DEM simulation.

3 Computational Setup

3.1 KCS Geometry

A modern container ship model, the KRISO Container Ship (KCS), was used to validate the numerical model for navigation in ice floes. KCS is a typical container ship model widely employed in computational simulations, which makes it valuable for comparison with existing reference data [7].

The full-scale length of the hull is 230 m, and a scale ratio of 1:52.667 is applied in this study, resulting in the following characteristic parameters:

- Length between perpendiculars: $L_{pp} = 4.367\,\mathrm{m}$
- Breadth: $B = 0.611\,\mathrm{m}$
- Draft: $T = 0.20489\,\mathrm{m}$
- Wetted surface area: $S = 3.483\,\mathrm{m}^2$

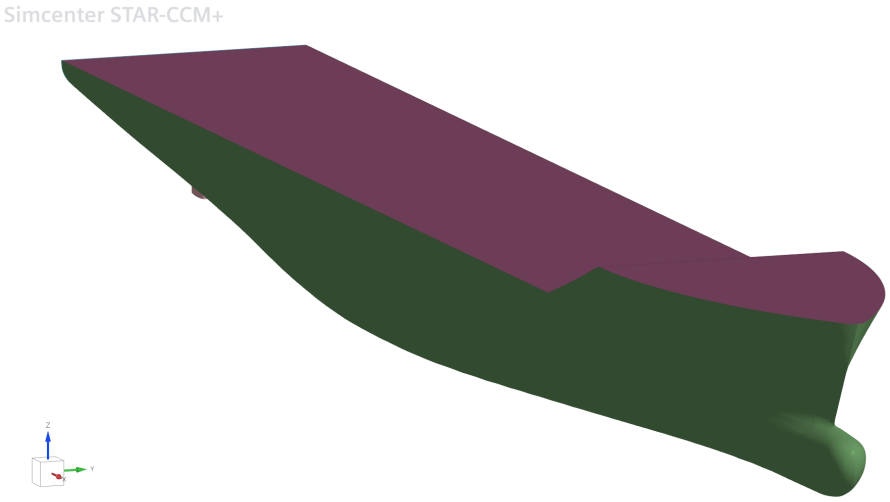


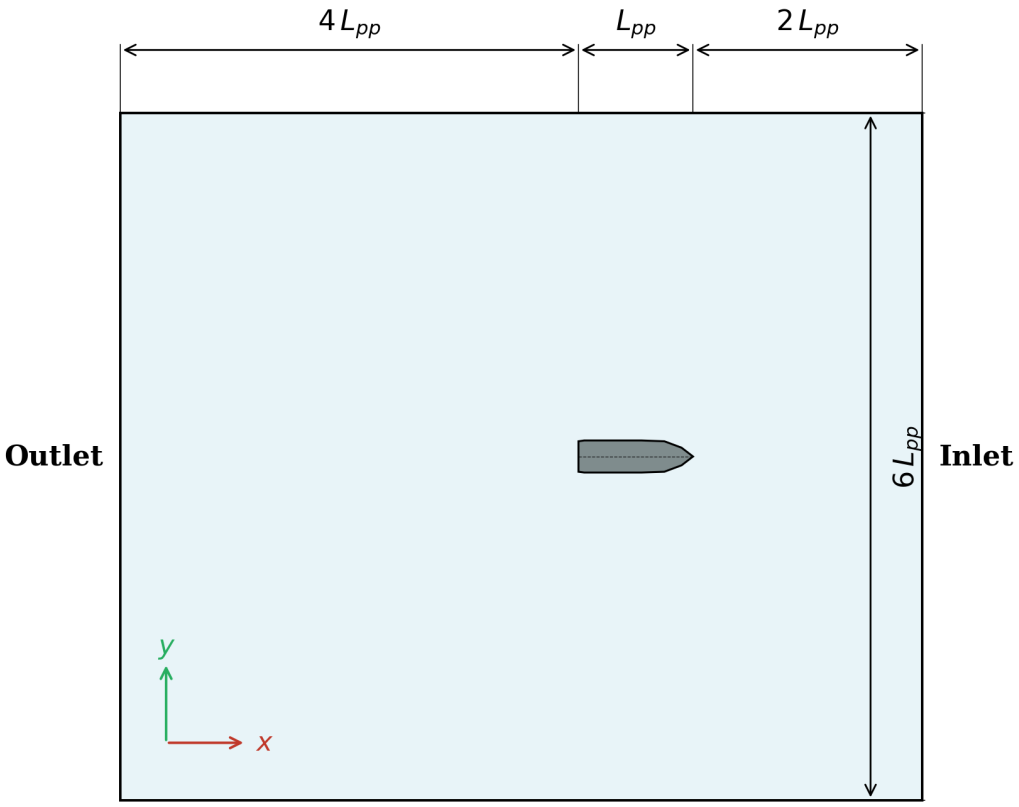
Figure 7: KCS geometry

3.2 Domain and Boundary Conditions

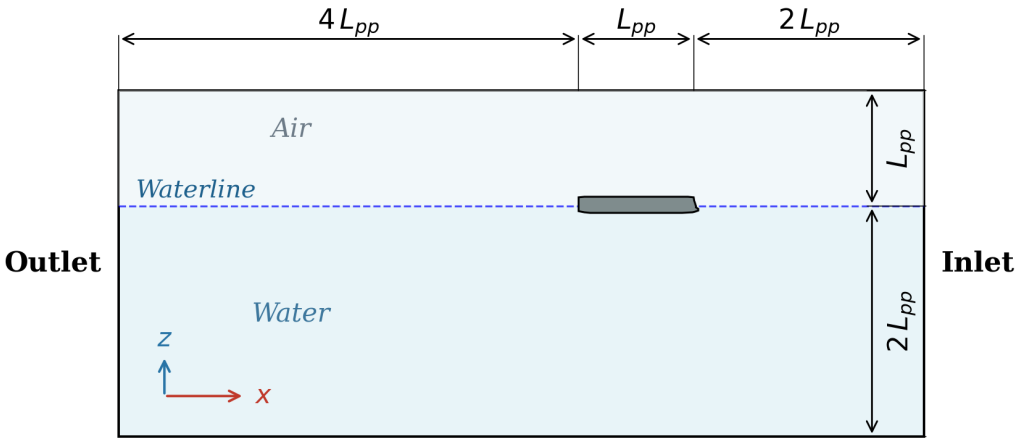
For the numerical simulation of the ship resistance in calm water, different approaches can be used. The ship can either be kept fixed while an incoming flow is imposed, or it can be allowed to move through an initially still fluid. In the present study of the KCS hull, the fixed-ship approach was selected, since it is simpler to implement from a numerical point of view. Moreover, for this type of resistance simulation, the expected differences between the two approaches are limited, as the motion is essentially steady and the acceleration effects remain small.

Following the guidelines of the International Towing Tank Conference (ITTC) [34], an open-ocean fluid domain was created, as illustrated in Figure 8. The computational domain is three-dimensional and defined in an earth-fixed Cartesian coordinate system Oxyz, where the (x,y) plane is parallel to the undisturbed free surface.

Unlike standard open-water CFD simulations, the full ship geometry must be included in the domain — rather than exploiting the longitudinal symmetry plane — since the random spatial distribution of ice floes breaks the port-starboard symmetry of the problem.



(a) Plan view (xy-plane)



(b) Profile view (xz-plane)

Figure 8: Computational domain of the KCS.

In this initial implementation, the ship is kept fixed with zero degrees of freedom (0 DOF). This simplification is justified at low advance speeds, where the vertical motion (heave) and pitch rotation induce only limited hydrodynamic effects, and allows a significant reduction in computational cost. The influence of ship motions on ice resistance will be investigated in future work.

Water and air are both assigned the same constant velocity ($U_{water} = U_{air}$) at the inlet of

the domain. Thus, a relative velocity exists between the ship and the water, corresponding to the real-life condition in which the ship navigates through calm water at a speed $U_{ship} = U_{water}$. In the following, the ship velocity will not be written explicitly, but rather expressed through the Froude number, defined as $F_r = \frac{U}{\sqrt{gL_{pp}}}$, where g is the gravitational acceleration and L_{pp} is the ship length between perpendiculars.

For the remaining domain boundaries, a fixed dynamic pressure is imposed at the outlet, symmetry plane conditions are applied on the side boundaries, and inlet conditions are prescribed at the top and bottom boundaries. The ship surface is modelled as a no-slip wall.

3.3 Mesh

A trimmed-cell mesh is generated using local refinement zones; a surface remesher is applied to refine the imported hull geometry. Prism layers are also included to model the boundary layer near the hull surface. Since a wall-function approach is used ($y^+ \approx 30$), an accurate representation of the boundary layer is essential, as it governs both the frictional resistance (wall shear stresses) and the pressure resistance (pressure distribution around the hull and flow separation).

The mesh base size is 0.0675m. To ensure an accurate resolution, several mesh refinement zones are required. An overview of the refinement zones is given below.

Table 6: Mesh refinement zones.

Refinement zone	Refinement direction	Cell size (m)	Percentage of base size
Free Surface	Z	0.0042	6.25%
Bulbous bow	X, Y, Z	0.0084	12.50%
Stern	X, Y, Z	0.0084	12.50%
Hull	X, Y	0.0338	50.00%
Hull	Z	0.0169	25.00%
Kelvin Wake	X, Y	0.0338	50.00%

3.3.1 Free surface

In any ship resistance case, it is very important to accurately resolve the air-water interface, which requires a refined free-surface mesh. In addition, for ice floes, it is recommended to use at least 4 cells per ice thickness , which corresponds to a cell size of 5 mm for an ice thickness of 20 mm.[15]

3.3.2 Kelvin Wake

These refinement zones are quite general for a mesh used in a ship resistance study in deep water. One difference is the Kelvin wake region, which is more refined to better track the particle motion[16].

3.3.3 Prism Layers

In the present study, to optimize computational efficiency, the viscous sublayer is not directly resolved. Instead, the mesh is designed such that the first cell height corresponds to the log-law region, targeting a value $y^+ \geq 30$.

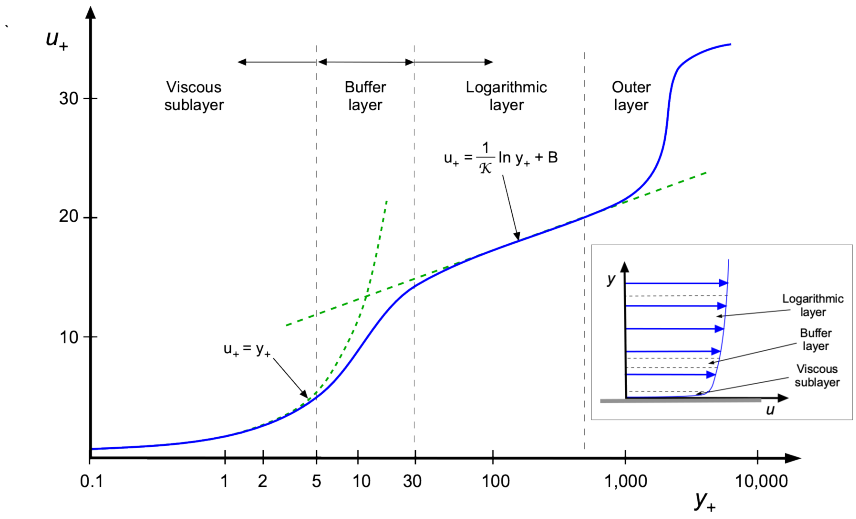


Figure 9: Turbulence regions in the boundary layer [35]

To define the near-wall mesh, the thickness of the inflation layer (prism layers) is determined based on the estimated boundary layer thickness, δ. In this study, the prism layer is designed to cover approximately 20% of the theoretical boundary layer thickness to ensure a smooth transition between the structured near-wall mesh and the unstructured far-field mesh. This value has been verified in CFD by assuring the boundary layer is within the prism layer region.

The first cell height, Δy, is calculated to target a dimensionless wall distance of $y^+ \geq 30$, consistent with the use of wall functions. This height is estimated using the following relations:

$$\Delta y = \frac{y^+ \mu}{\rho u_\tau} \tag{45}$$

where the friction velocity u_τ is derived from the wall shear stress τ_w:

$$u_\tau = \sqrt{\frac{\tau_w}{\rho}} = U_\infty \sqrt{\frac{C_f}{2}} \tag{46}$$

The skin friction coefficient C_f is estimated using the ITTC-1957 correlation line for ship-flow applications [34]:

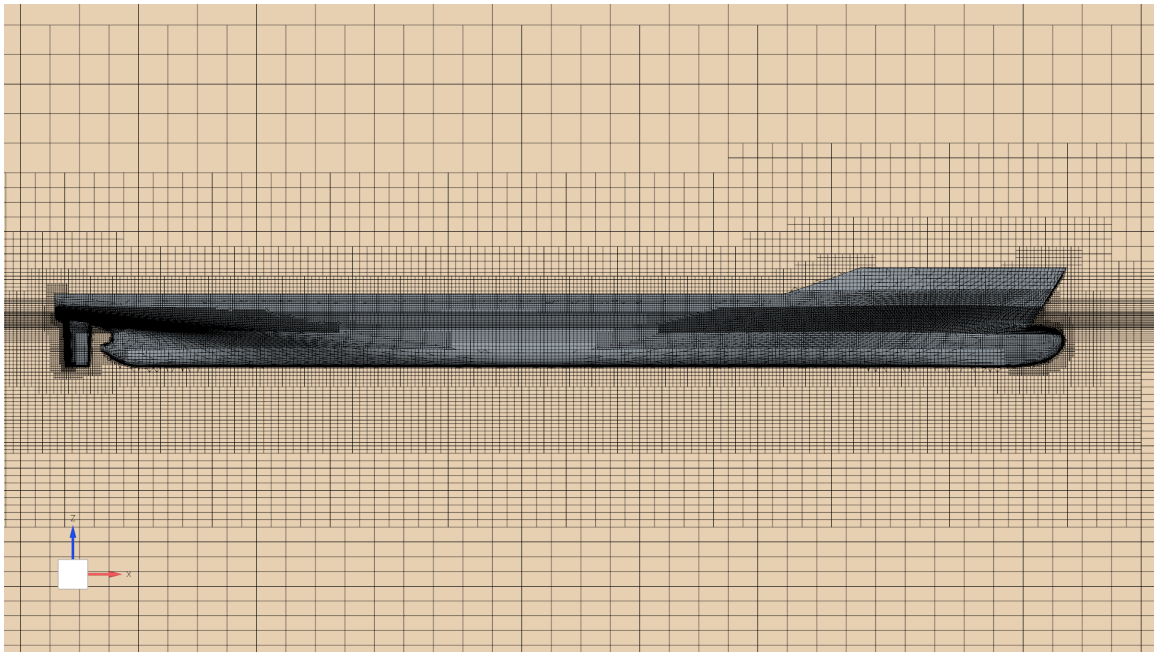
$$C_f = \frac{0.075}{(\log_{10} Re - 2)^2} \tag{47}$$

3.4 Selection of CFD Numerical Parameters

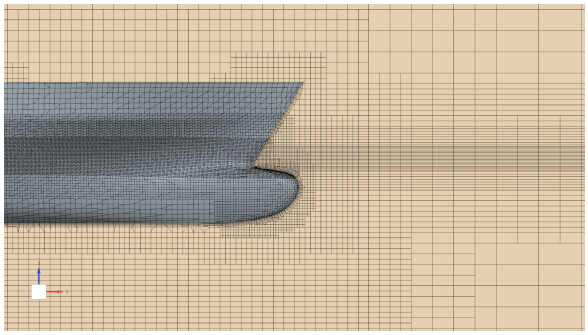
For all the simulations, the time step has been set to $dt = 0.01\,\text{s}$.

The convergence within each time step was controlled by the number of **inner iterations**. Preliminary tests were conducted to determine the optimal balance between stability and computational cost:

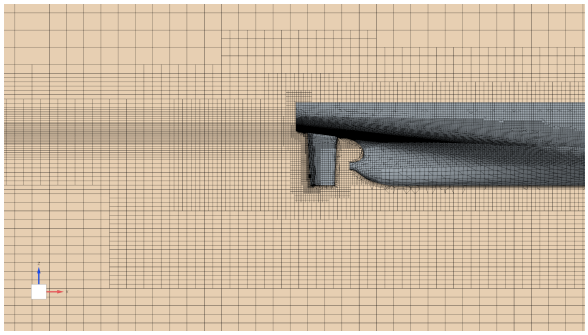
- Five inner iterations were found to provide a robust compromise, ensuring stable convergence of the residuals at each step.
- Increasing this number beyond five yielded no significant improvement in solution accuracy, whereas lower values led to unstable behavior and poor pressure-velocity coupling.



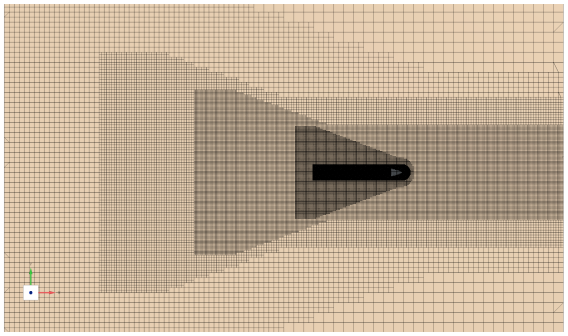
(a) Side view of the mesh around the hull.



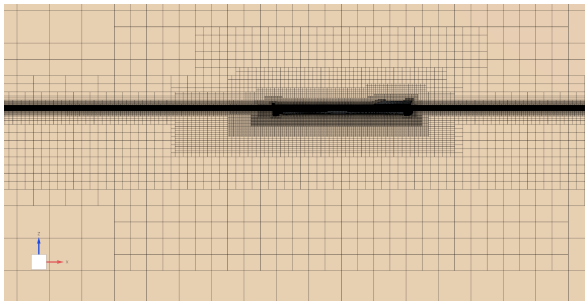
(b) Detailed view of the mesh at the bow.



(c) Detailed view of the mesh at the stern.



(d) View at the free surface.



(e) View on the symmetry plane.

Figure 10: Numerical mesh of the computational domain and local refinements around the geometry.

3.5 Selection of DEM/Lagrangian Numerical Parameters

3.5.1 Material properties

The ice floes are defined as solid, cylindrical DEM particles with constant material properties. The ice density is set to $\rho_{\text{ice}} = 900.0\,\text{kg/m}^3$, while the Poisson's ratio and Young's modulus are set to $\nu_{\text{ice}} = 0.33$ and $E_{\text{ice}} = 9.0$ GPa, respectively [20].

For the contact model, different friction coefficients are assigned depending on the type of interaction. The ice–ice friction coefficient is set to 0.35, while the ice–ship friction coefficient is set to 0.035 [7]. The same restitution coefficient is used for both the normal and tangential directions, and for both ice–ice and ice–steel contacts, with $C_{\text{rest}} = 0.8098$.

3.5.2 Ice floe Geometry and Size Distribution

The geometry selected to represent the ice floes in the simulation is a cylinder. Although ice floes are modelled as square prisms in the experimental reference cases, the cylindrical shape was preferred for two reasons: the influence of floe geometry on the added resistance has been shown to be limited [16], and the cylindrical shape considerably simplifies the implementation. To ensure a consistent size equivalence between the experimental and numerical ice floes, both geometries are matched by cross-sectional area: for a given floe size probability, a simulated cylindrical floe and its experimental square counterpart share the same surface area, and consequently the same mass.

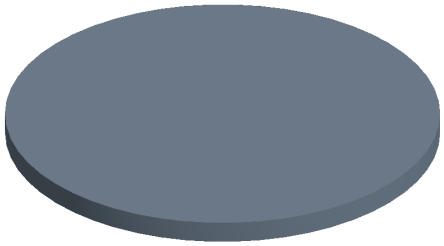


Figure 11: Example of ice floe: Diameter = $0.4m$, thickness = $0.02m$

The ice floe diameters are assumed to follow a log-normal distribution [19].

$$f_X(x) = \frac{1}{x\sigma_{\text{lg}}\sqrt{2\pi}} \exp\left(-\frac{(\ln x - \mu_{\text{lg}})^2}{2\sigma_{\text{lg}}^2}\right) \tag{48}$$

For the KCS case, the parameters of the log-normal distribution were selected to match the experimental data reported in [7].

The resulting diameter probability density function is shown in Figure 12.

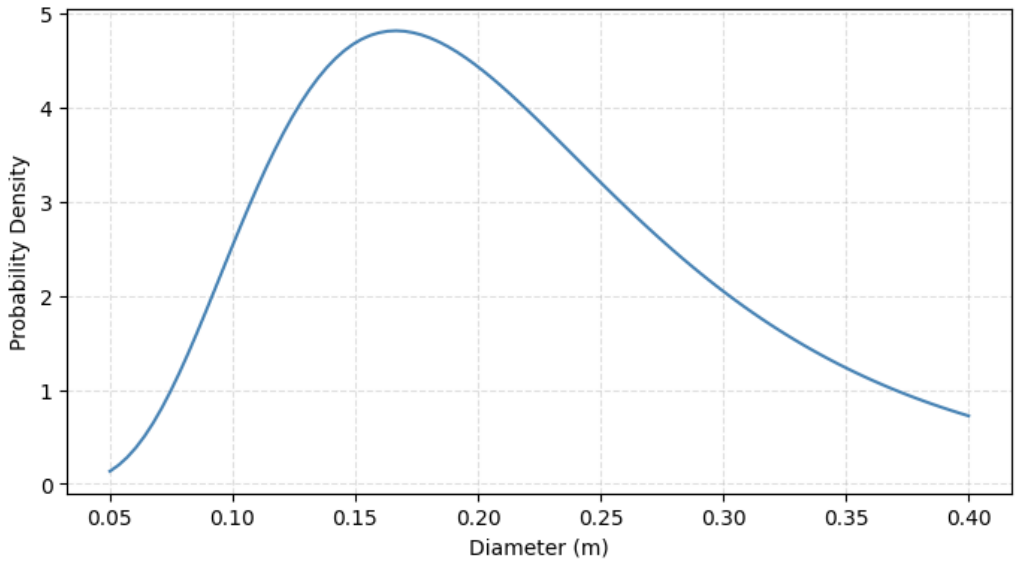


Figure 12: Ice floes diameter distribution: log-normal, $\mu_{\text{lg}} = -1.59$, $\sigma_{\text{lg}} = 0.45$

The ice floe diameters range from 5 cm to 40 cm with a constant thickness $t_{\text{ice}} = 2\,\text{cm}$.

3.5.3 Ice Floe Field Generation

The number of ice floes required to reach a prescribed ice concentration is determined from the cumulative ice-covered area. Ice floes are generated with random diameters drawn from the selected log-normal distribution until the total ice area exceeds:

$$c_{\text{ice}}\, A_{\text{domain}},$$

where c_{ice} is the target ice concentration and A_{domain} is the surface area of the ice domain. Two algorithms were written to generate the ice floe field.

3.5.4 First Algorithm: Random Placement

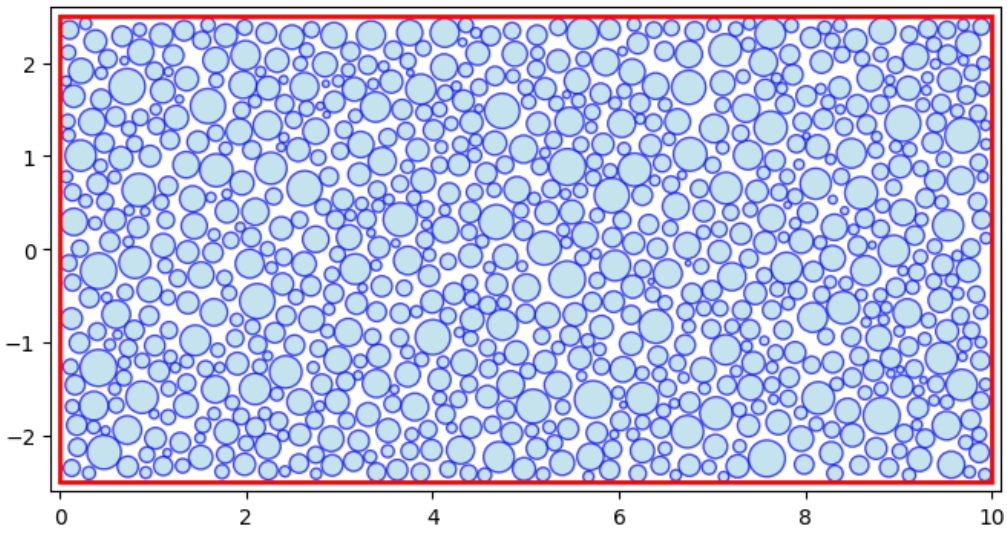


Figure 13: Example of ice floes field generation with the first algorithm, $c_{\text{ice}} = 60\%$

A first simple algorithm was developed. In this approach, the floes are sorted in decreasing order of diameter and then placed randomly inside the domain while avoiding overlap (See Figure 13). This method provides satisfactory results for ice concentrations below or equal to 60%. However, for higher concentrations, the algorithm becomes less efficient, since the available open-water space decreases significantly and the random placement of all floes becomes increasingly difficult. The code is available in Appendix A.1

3.5.5 Second Algorithm: Progressive Growth

A second algorithm was implemented in order to generate ice floe fields with high ice concentration. This method is based on a progressive-growth strategy. First, the final floe diameters are generated from a truncated log-normal distribution, bounded by a minimum and maximum diameter. The number of floes is chosen such that the target ice concentration is reached, where the concentration is defined as

$$c_{\text{ice}} = \frac{\sum_{i=1}^{N} \pi r_i^2}{S_{\text{ice}}},$$

with r_i the radius of floe i, N the total number of floes, and S_{ice} the surface area of the ice domain.

Instead of placing the floes directly at their final size, all floes are initially inserted with a reduced radius (45% of their final radius). This substantially reduces the probability of initial overlap and makes the packing process more robust. The initial positions are chosen randomly inside the domain using an empty-space criterion: for each floe, several candidate positions are tested, and the position with the largest clearance from the already placed floes is selected.

After this initial placement, the floes are progressively enlarged until their final size is reached. At each growth step, the radius of every floe is multiplied by a scale factor that is gradually increased from the initial value to 1. After each increase in size, the algorithm checks the overlap between all pairs of floes. For two floes i and j, the overlap is defined as

$$\delta_{ij} = r_i + r_j - \|\mathbf{x}_i - \mathbf{x}_j\|,$$

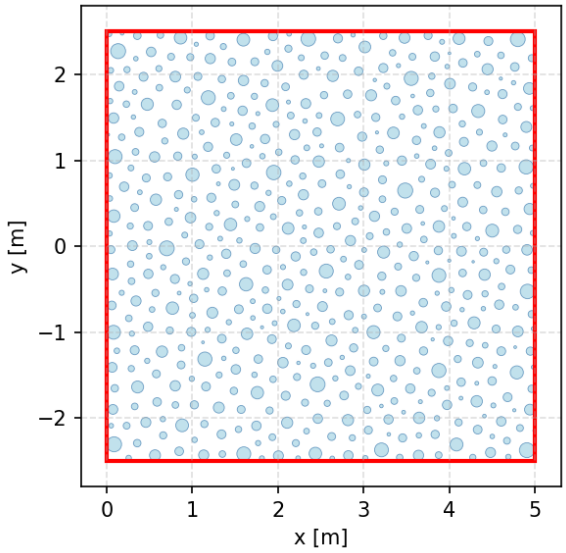
where $\mathbf{x}_i$ and $\mathbf{x}_j$ are the floe center positions. An overlap occurs when $\delta_{ij} > 0$. The global overlap penalty is then computed as

$$P = \sum_{\delta_{ij}>0} \delta_{ij}^2.$$

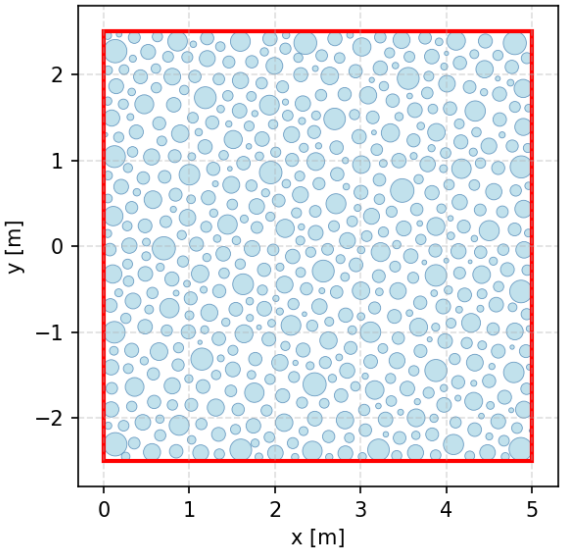
If overlaps are detected, a relaxation procedure is applied. Overlapping floes are pushed apart in the direction connecting their centers. Smaller floes are allowed to move more than larger floes, which improves the stability of the packing. If the overlap penalty remains too large after relaxation, the growth step is rejected and reduced. In difficult configurations, some of the smallest problematic floes are relocated to better positions in order to escape local blocked arrangements.

Once the final scale is reached, a final relaxation step is performed. If only very small residual overlaps remain, a slight uniform reduction of all floe radius is applied to remove them completely while keeping the ice concentration almost unchanged. This correction is negligible compared with the floe size, since it is only applied when the maximum overlap is below a prescribed tolerance.

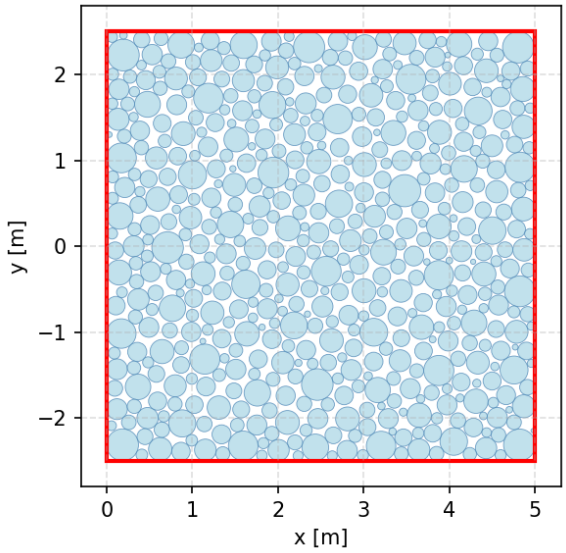
This progressive-growth algorithm is efficient for high-concentration ice fields. In the present case, it allows the generation of an ice floe field with a target concentration of $c_{\text{ice}} = 80\%$ in a domain of area $S_{\text{ice}} = 25$ m^2. The different stages of the growth process are shown in Figure 14. The code is available in Appendix A.2.



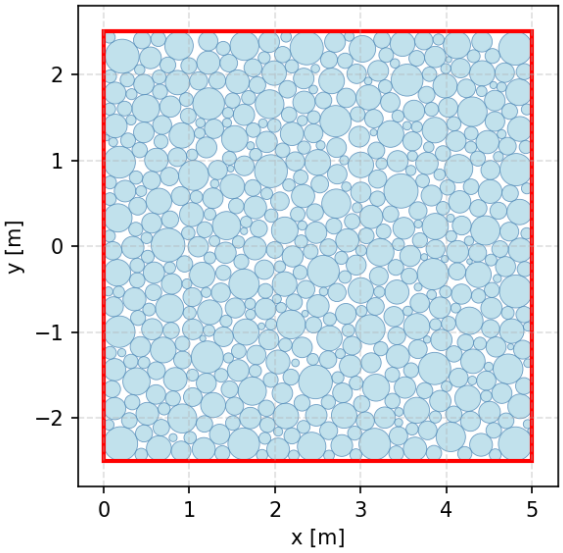
(a) Progressive growth $n°0$, scale = 0.440, c_{ice} = 15.5%



(b) Progressive growth $n°5$, scale = 0.686, c_{ice} = 37.7%



(c) Progressive growth $n°10$, scale = 0.936, c_{ice} = 70.2%



(d) Final progressive-growth ($n°20$) ice floe field, c_{ice} = 80.1%

Figure 14: Example of generation of ice floe field with Algorithm II, $c_{\text{ice}} = 80.1\%$, $S_{\text{ice}} = 25\,\text{m}^2$, 538 floes.

3.5.6 Injection of the ice floes

The introduction of ice floes into the computational domain is performed using an array injection strategy. The simulation is first run without ice floes in order to allow the fluid solution to reach a steady-state condition. Once this state is achieved, an array of ice floes is injected into the domain near the inlet. Each floe is initialized with a velocity equal to the fluid velocity. When the first array has travelled a distance L_{array}, a second array is injected so that it follows the previous one. This process is repeated to ensure a continuous inflow of ice floes into the computational domain.

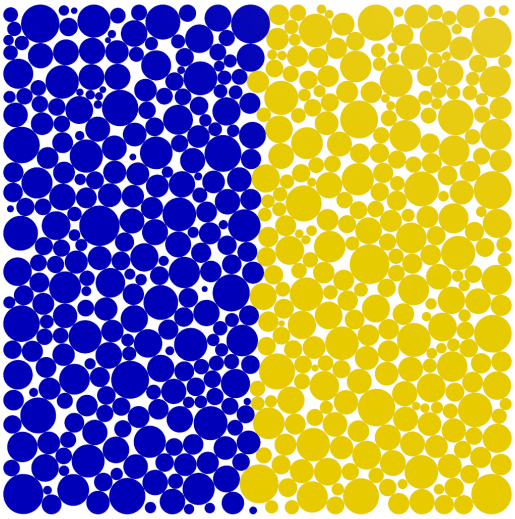


Figure 15: Illustration of how the ice floes are injected: in blue the first ice array (at $t = t_1$), in yellow the second ice array (at $t = t_1 + \Delta t_{\text{inject}}$)

3.6 Sensitivity Test on DEM parameters

See Section 2.2.3 for details on these parameters.

Several sensitivity tests were performed to choose the appropriate values for the DEM time scale and the spring stiffness constant K.

The aim is to identify parameter values that limit the maximum inter-particle overlap to approximately 1mm while maintaining an acceptable computational cost.

3.6.1 Setup of the Sensitivity Test

The setup is simple: a domain with no gravity and no water is created, with a wall. A DEM particle is launched toward the wall at a fixed velocity, representing the ice floe with the maximum diameter used in the experiments. The maximum overlap at the wall, the maximum DEM force, the DEM force impulse, and the CPU time are then recorded.

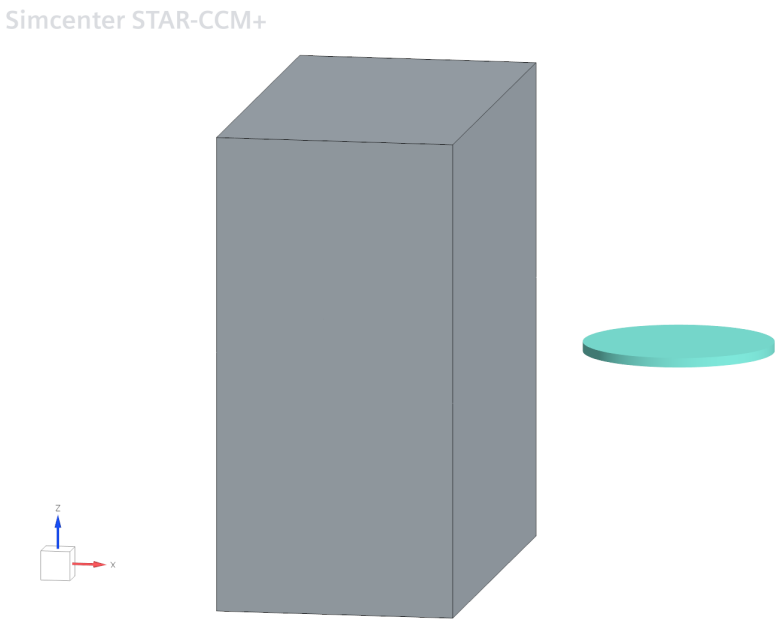


Figure 16: Setup for the sensitivity test.

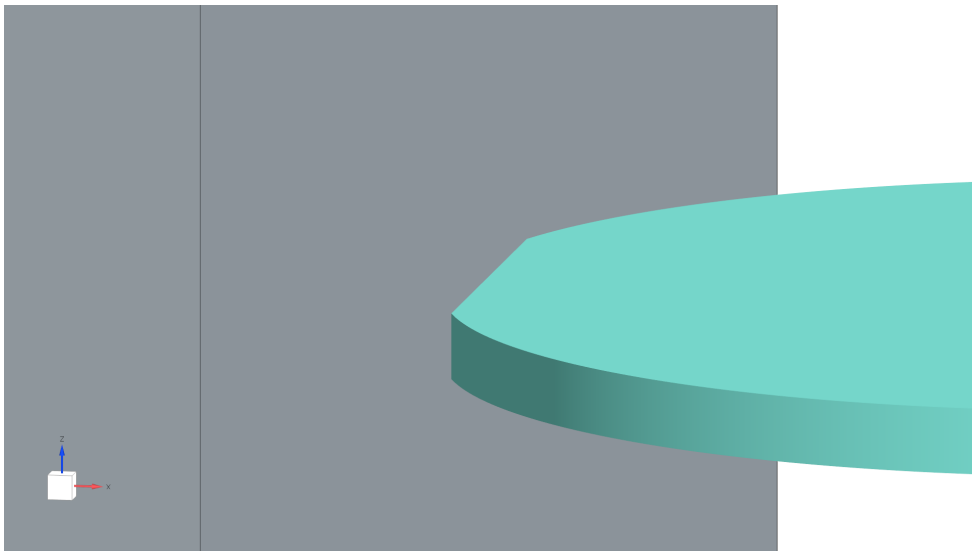


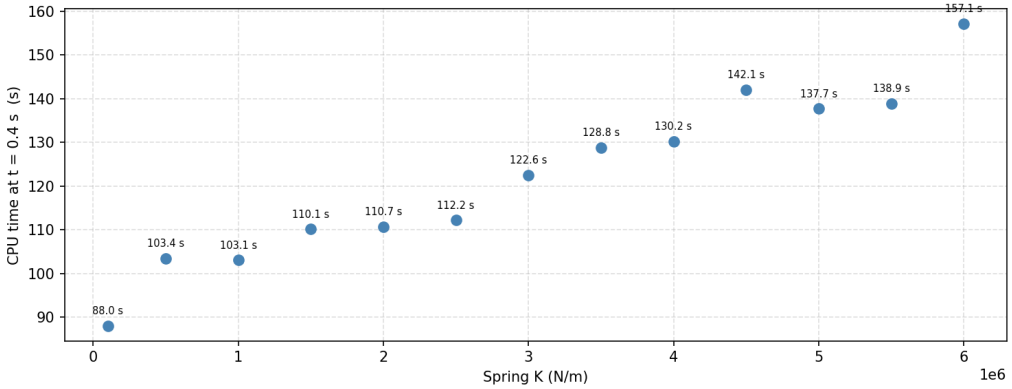
Figure 17: Example of overlap between the wall and the ice floe.

3.6.2 Spring Stiffness Constant K

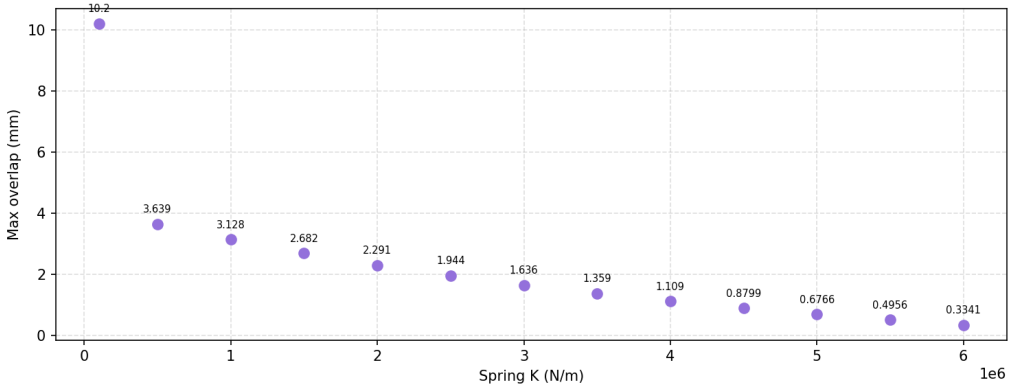
For this test, the time scale was set to 0.4 and the CFD timestep to $dt = 0.01\,\text{s}$. The spring stiffness K was then tested from 0.1 to 6 MN/m, and each simulation was run for 0.4 s.

The selection of K involves a trade-off between two competing requirements. A high K gives more accurate results, especially for the DEM force impulse, and reduces the overlap. But a high K also makes the DEM time step smaller (see Equation (23)), which significantly increases CPU time.

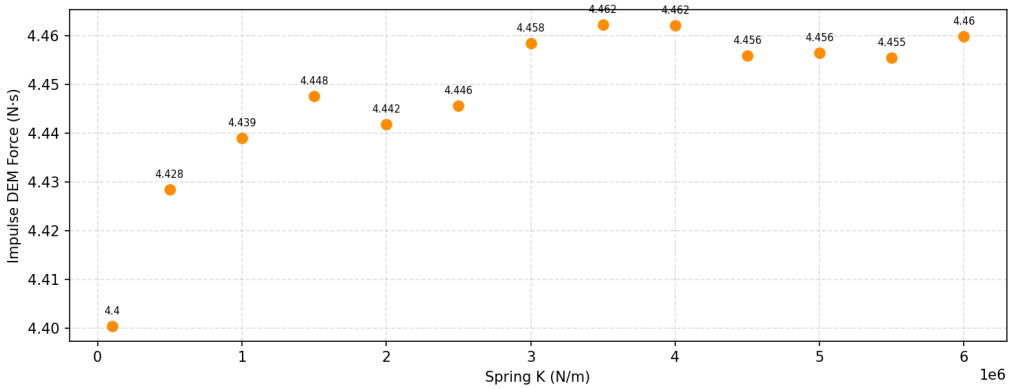
The value $K = 4\,\text{MN/m}$ was chosen because it gives good results for the DEM force impulse, a small enough overlap, and a reasonable CPU time.



(a) CPU time.



(b) Maximum overlap (mm).



(c) DEM force impulse (N·s).

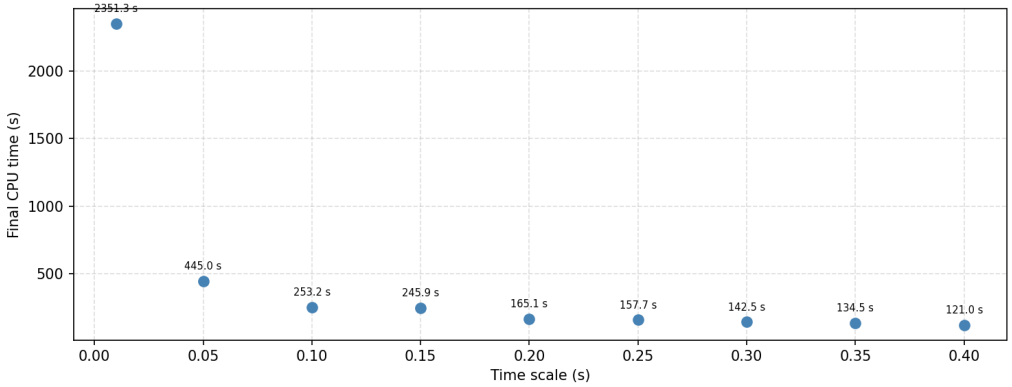
Figure 18: Sensitivity test results for the spring stiffness constant K.

3.6.3 Time Scale

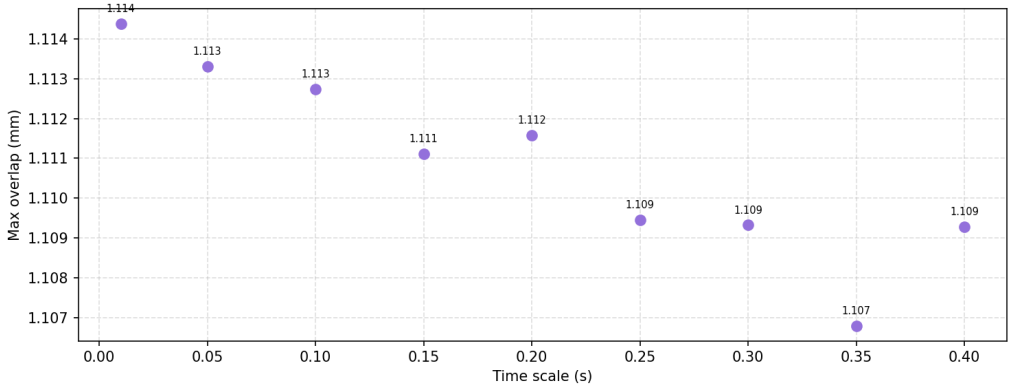
The effect of the time scale t_{scale} was studied by looking at the DEM force impulse, the maximum overlap, and the CPU time. The time scale is used to set the DEM time step in the solver. STAR-CCM+ recommends values between 0.1 and 0.4, but some documentation suggests using values as low as 0.05 or 0.01 [22]. The range from 0.01 to 0.4 was therefore tested, with $dt = 0.01\,\text{s}$ and $K = 4\,\text{MN/m}$.

The results show that the DEM force impulse and the maximum overlap are almost not

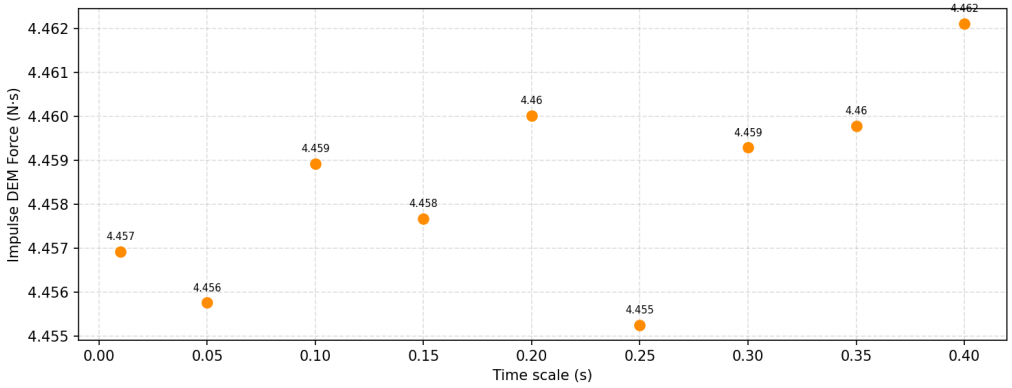
affected by the time scale in this study, with changes below 0.6%. However, the CPU time largely increases when the time scale is reduced. For this reason, $t_{\text{scale}} = 0.4$ was chosen to keep the CPU time as low as possible. a value greater than 0.4 was not chosen in order to remain within the recommendations of STAR-CCM+.



(a) CPU time.



(b) Maximum overlap (mm).



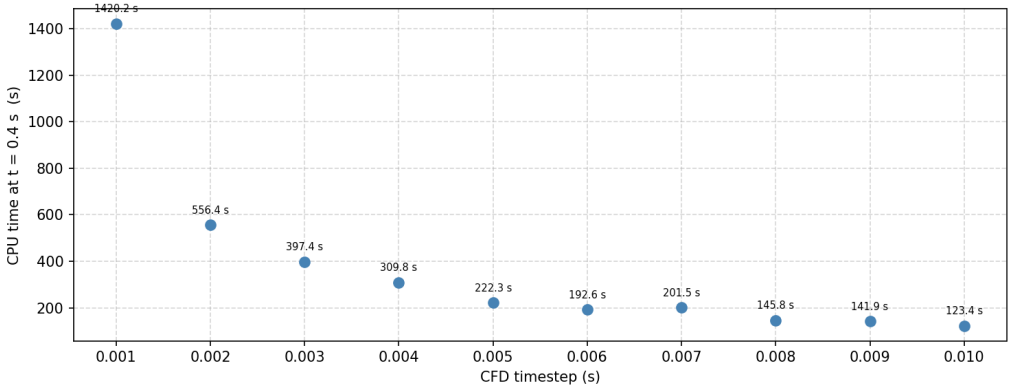
(c) DEM force impulse (N·s).

Figure 19: Sensitivity test results for the time scale t_{scale}.

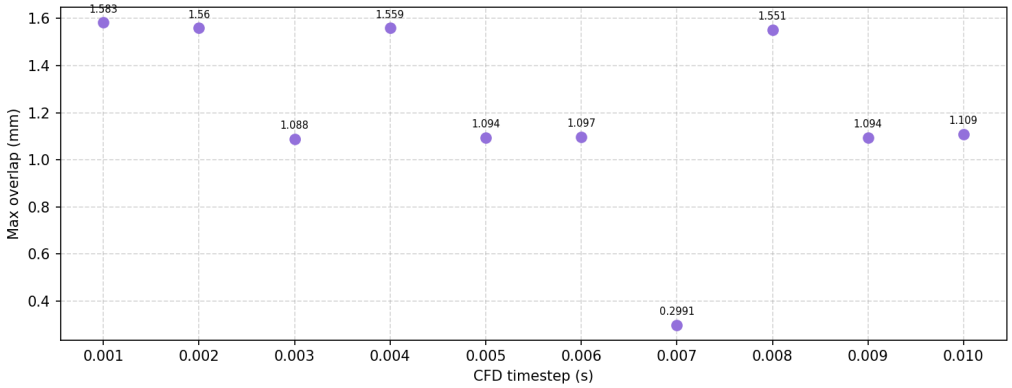
3.6.4 CFD Time Step

A last sensitivity test was done on the CFD time step, with $K = 4\,\mathrm{MN/m}$ and $t_{\mathrm{scale}} = 0.4$. Values from 0.001 s to 0.01 s were tested.

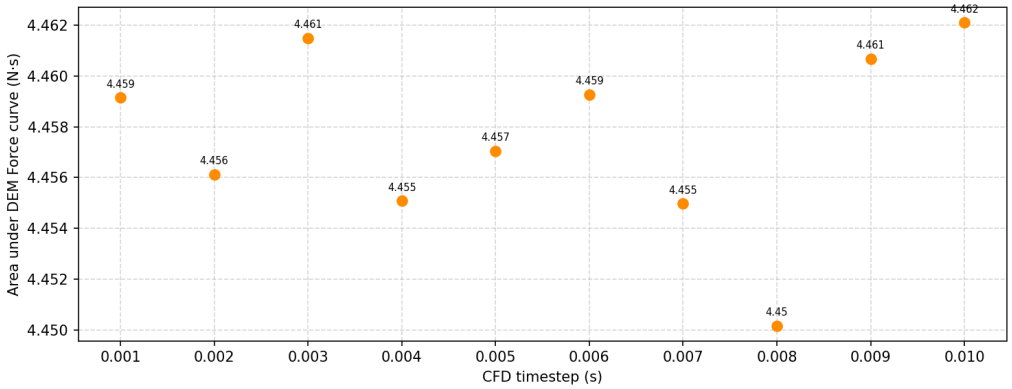
The results show that the CFD time step has very little effect on the DEM force impulse or the maximum overlap, with differences below 1%. Reducing the CFD time step increases the CPU time. Since $dt = 0.01\,\mathrm{s}$ is the time step used in the deep-water CFD study, this value was kept.



(a) CPU time.



(b) Maximum overlap (mm).



(c) DEM force impulse (N·s).

Figure 20: Sensitivity test results for the CFD time step dt.

3.6.5 Conclusion

With a spring stiffness $K = 4\,\mathrm{MN/m}$, a time scale $t_{\mathrm{scale}} = 0.4$, and a CFD time step $dt = 0.01\,\mathrm{s}$, the maximum overlap is close to 1 mm, and the DEM force impulse reaches $4.462\,\mathrm{N\cdot s}$, compared to a theoretical value of $4.5\,\mathrm{N\cdot s}$, which represents a relative error of $0.89\,\%$. Furthermore, these parameter values significantly reduce the computational time compared to recommended values from literature [22].

4 Results

Simulations were carried out to compare the numerical results with the experimental data reported in [7]. Five different ship speeds were considered, namely $V = 0.4$, 0.6, 0.8, 1.0, and 1.2 m/s, corresponding to Froude numbers of $F_r = 0.06$, 0.09, 0.12, 0.15, and 0.18, respectively. For each case, an ice floe field was injected over a duration of 40 s. Such a simulation time is required because the contacts between the hull and the ice floes generate force peaks. Therefore, the resistance must be averaged over a sufficiently long time interval in order to obtain representative mean values.

4.1 Added Ice Resistance

4.1.1 Results and comparison with experimental results

The resistance prediction results from the simulations are compared with the experimental results from Guo et al. [7].

The simulation results take the form of spikes due to the contact of the ice floes on the ship hull, as shown in Figure 21(a).

To better visualise the DEM forces acting on the hull, it is more convenient to work with the DEM impulse (unit: N·s) rather than the raw force signal. This approach serves two purposes: it allows one to check whether the simulation has run long enough to reach convergence, and it gives a reliable estimate of the mean DEM force on the ship.

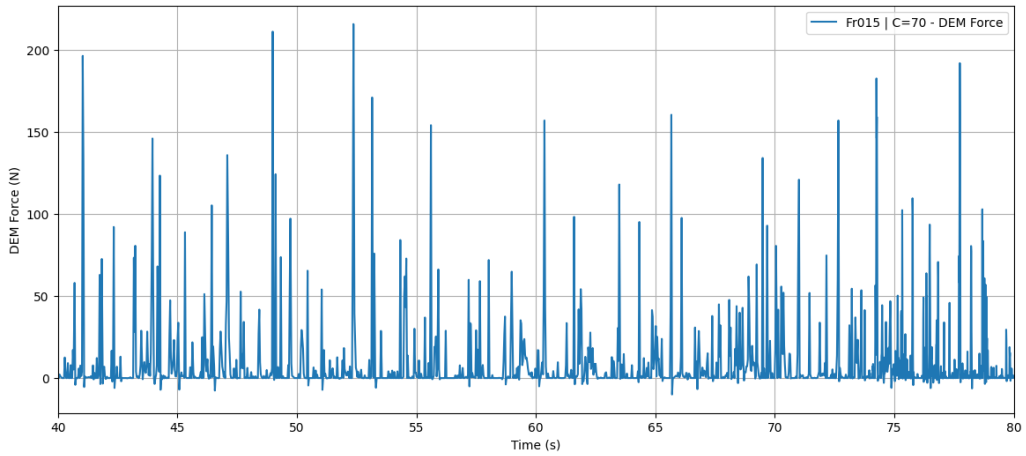
The cumulative DEM impulse is the area under the DEM force curve over time, as shown in Figure 21(b), defined as:

$$I(t) = \int_0^t F_{\mathrm{DEM}}(\tau)\, d\tau \tag{49}$$

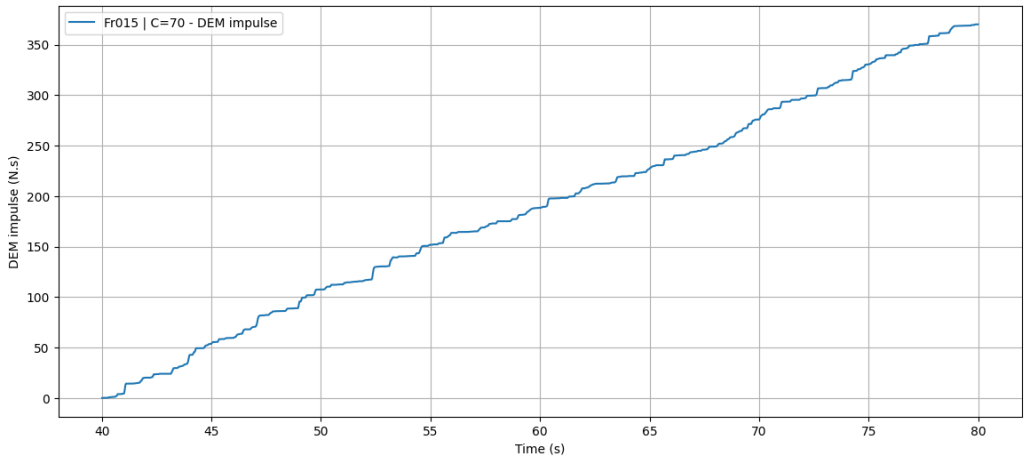
The cumulative ice resistance is then obtained by dividing the cumulative impulse by the elapsed time:

$$R_{\mathrm{ice}}(t) = \frac{I(t)}{t} \tag{50}$$

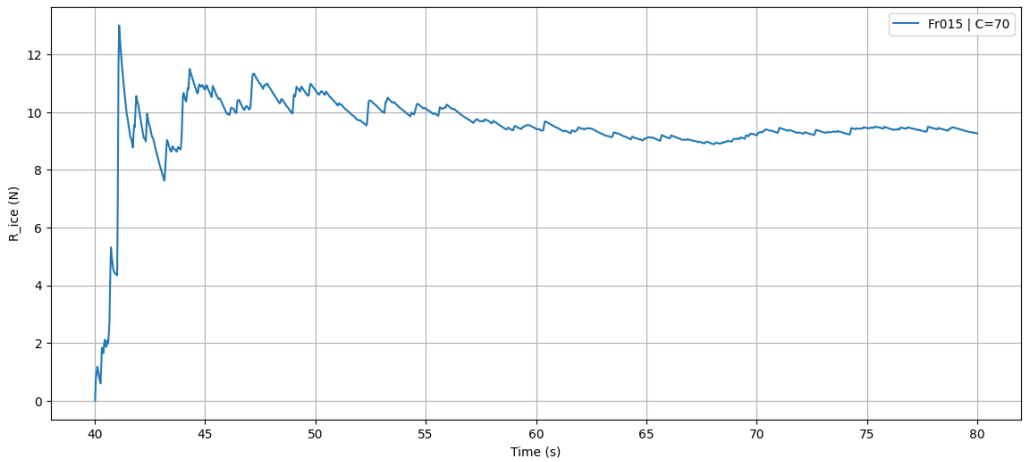
When this quantity stabilises and converges to a constant value, it confirms that the simulation has run long enough and that the mean ice resistance can be read directly from the plateau. As shown in Figure 21c, convergence is reached after approximately 40 seconds, which validates the chosen simulation duration.



(a) Raw DEM force on the hull over time. The spikes correspond to individual ice floe contacts.



(b) Cumulative DEM impulse $I(t)$, representing the time-integrated DEM force on the hull.

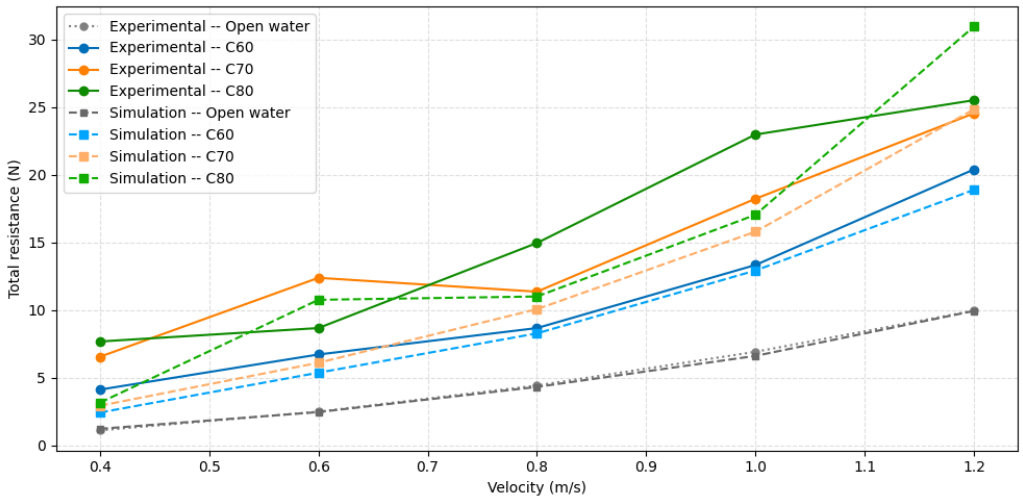


(c) Cumulative ice resistance $R_{\text{ice}}(t) = I(t)/t$. Convergence to a constant value confirms that the simulation duration is sufficient.

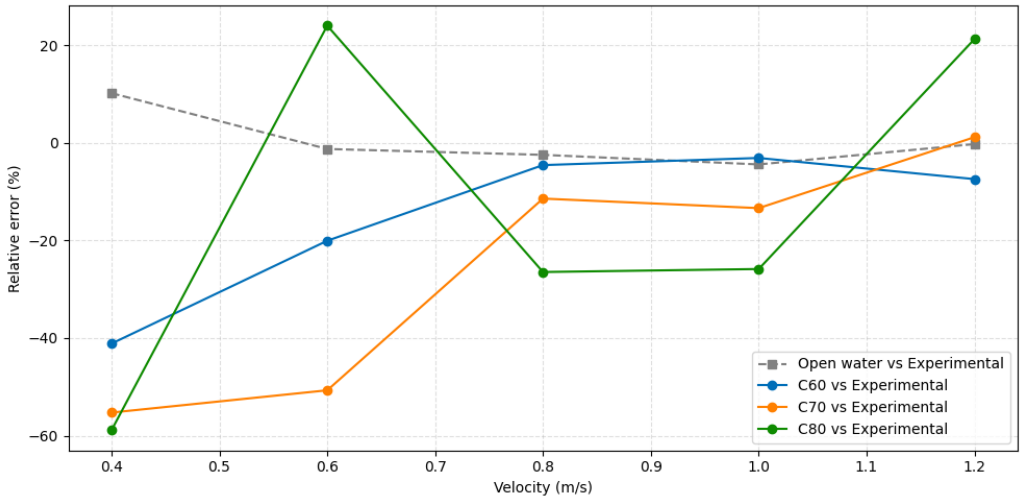
Figure 21: DEM force on the ship hull (a), cumulative DEM impulse (b), and convergence of the mean ice resistance (c).

Simulations were run for all velocities at ice concentrations of 60%, 70% and 80%. The total

resistance values are compared with the experimental results in Figure 22 and Figure 23.



(a) Experimental and simulated total resistance for $c_{\text{ice}} = 60\%$, 70% and 80%.

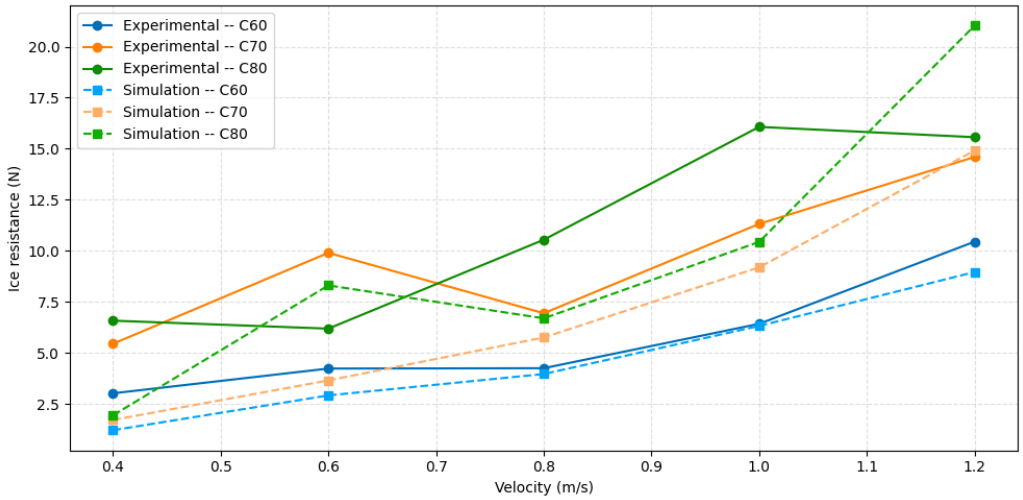


(b) Relative error $\frac{R_{sim}-R_{exp}}{R_{exp}} \times 100$ between experimental and simulated total resistance.

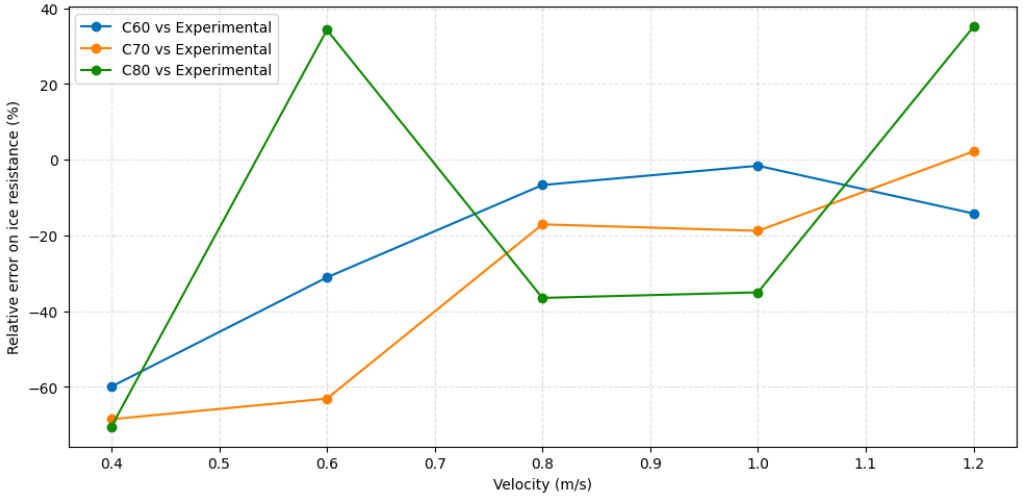
Figure 22: Comparison of total resistance between simulation and experiment (a), and corresponding relative error (b).

Velocity	Open Water	$c_{\text{ice}} = 60$	$c_{\text{ice}} = 70$	$c_{\text{ice}} = 80$
0.4	+10.23%	−41.13%	−55.28%	−58.89%
0.6	−1.21%	−20.03%	−50.71%	+24.08%
0.8	−2.42%	−4.52%	−11.39%	−26.45%
1.0	−4.28%	−4.55%	−14.44%	−26.70%
1.2	−0.21%	−7.39%	+1.21%	+21.33%

Table 7: Relative error between experimental and simulated total resistance $\frac{R_{sim}-R_{exp}}{R_{exp}} \times 100$



(a) Experimental and simulated Ice resistance for $c_{\text{ice}} = 60\%$, 70% and 80%.



(b) Relative error $\frac{R_{\text{ice,sim}} - R_{\text{ice,exp}}}{R_{\text{ice,exp}}} \times 100$ between experimental and simulated ice resistance.

Figure 23: Comparison of ice resistance between simulation and experiment (a), and corresponding relative error (b).

First, similar behaviors are observed for $c_{\text{ice}} = 60\%$ and 70%: the simulations tend to underestimate the resistance (both the total resistance and the ice resistance alone). At higher velocities (above 0.8 m/s), the agreement with the experimental data is relatively good, with relative errors below 15% on the total resistance. However, at lower velocities (0.4 and 0.6 m/s), the relative error increases significantly, reaching values between 20% and 60%. A possible explanation for these large inconsistencies at low speed is discussed in Section 4.2.2.

At higher ice concentration ($c_{\text{ice}} = 80\%$), the simulations tend to underestimate for 0.8 and 1.0 m/s but overestimate for 1.2 m/s, with relative errors approaching 25% on the total resistance. Hypotheses about these errors are discussed in Section 4.2.3.

Another noteworthy result is the resonance-like behavior observed in the simulation at $V = 0.6$ m/s for an ice concentration of $c_{\text{ice}} = 80\%$. A similar phenomenon was documented in the experimental campaign of Guo et al. [7], also at $V = 0.6$ m/s, but for a lower concentration of $c_{\text{ice}} = 70\%$. At this specific velocity, instead of pushing the ice floes aside, the ship model carries them along with it, causing an abrupt increase in total resistance:

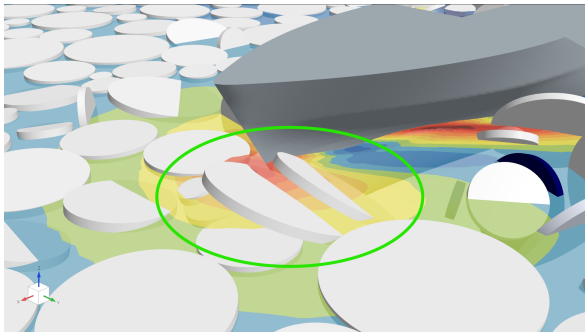
> "In the special region around 0.6 m/s, the frequency of the ship model encountering the pack ice occasionally shows a certain equilibrium between a few factors, namely the friction between the body and the ice, and the water resistance on the ice. Hence, instead of pushing the ice aside, the ship model brings the ice into motion along with it, resulting in an abrupt increase in the total resistance as compared with the total resistance at 0.4 m/s. The total resistance reaches its maximum around 0.6 m/s." [7]

The simulation reproduces this mechanism at the same critical velocity, lending credibility to the numerical approach. While the resonance is triggered at $c_{\text{ice}} = 70\%$ in the experiment and at $c_{\text{ice}} = 80\%$ in the simulation, both cases exhibit the abrupt resistance peak at $V = 0.6$ m/s, confirming that the simulation captures this specific ice–ship interaction behavior.

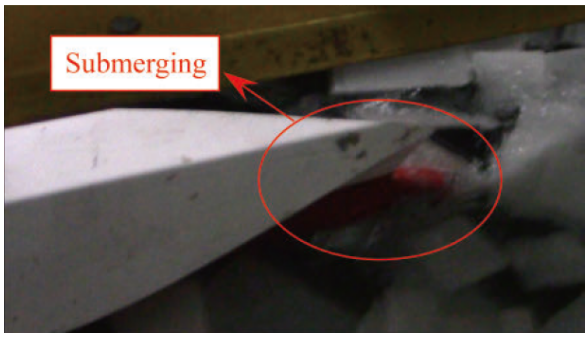
4.2 Ice Floe Behavior upon Contact with the Ship

4.2.1 Rotating, Submerging and Sliding

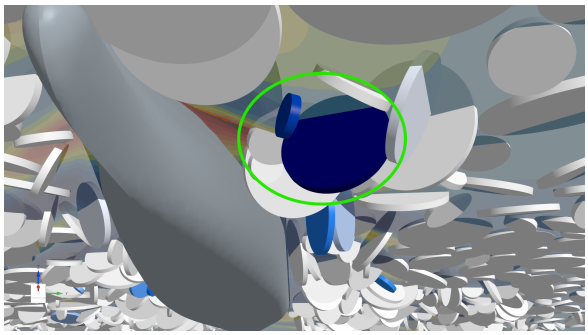
(a) Rotating (experiment)



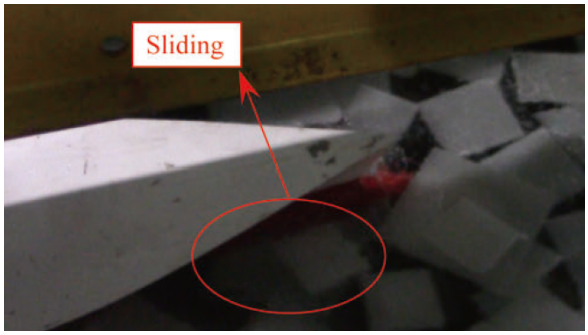
(b) Rotating (simulation)



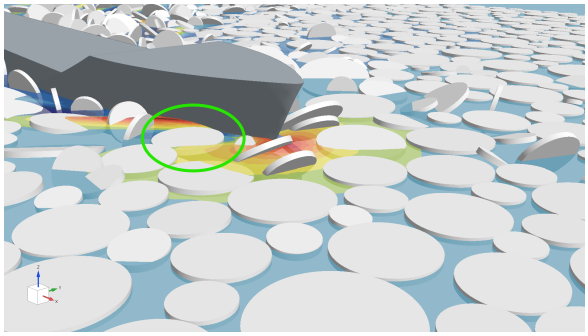
(c) Submerging (experiment)



(d) Submerging (simulation)



(e) Sliding (experiment)



(f) Sliding (simulation)

Figure 24: Rotating, submerging and sliding of ice floes: experimental results [7] (left) and simulation in STAR-CCM+ (right).

Different behaviors can be observed when the ice floes come into contact with the ship. From experimental results [7], three main behaviors can be identified: rotating, submerging, and sliding, that can be also observed in the simulations.

4.2.2 Non-Realistic Behavior at Low Velocity

As explained in Section 2.2.5, the ice floes are represented as DEM particles, a simplification that neglects several of the hydrodynamic forces exerted by the fluid. Scaling the moment of inertia and adding a drag torque artificially restores some of these missing effects, but others remain absent, in particular the hydrostatic restoring moment that tends to keep a floating floe horizontal.

This limitation has a direct consequence on the simulated ship–ice interaction. Guo et al. [7] observed that at low speed (around 0.4 m/s) the floes are only slowly pushed aside by the hull, whereas at higher speed (around 1.0 m/s) they are pushed aside more abruptly and tend to rotate and submerge. At low speed, in other words, the floes remain essentially horizontal, with the axis of the cylindrical particles staying vertical. In the present simulations, however, the absence of the restoring moment means that the slightest contact with the hull is enough to make a floe rotate. This premature rotation interrupts the chain reaction through which the floes are normally pushed against one another: the upstream floes no longer transmit their thrust to the floes in contact with the hull, which leads to an underestimation of the ice resistance.

Overall, the ice floes should mostly slide along the hull (Figure 24f), whereas in the current simulations they rotate (Figure 24b) and submerge (Figure 24d) more frequently than observed experimentally. One way to improve the model would be to change the particle model. See Section 5.

4.2.3 Non-Realistic Behavior at High Concentration

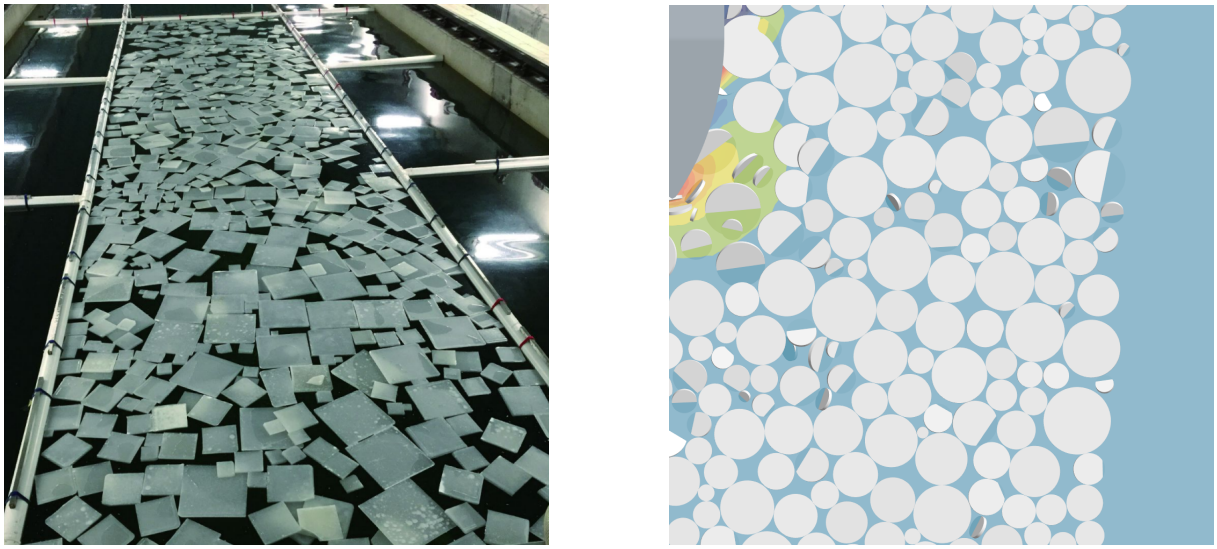
At 80% ice concentration, significant errors are observed in the ice resistance predictions. Three possible hypotheses are discussed below.

First, the shape of the ice floes may play a role. Cylindrical floes were selected on the grounds that shape should not significantly affect the results [16]; however, this assumption was validated only up to a concentration of 70%, not 80%. Circular ice floes allow lower maximum surface coverage than square ones for the same size distribution, since circles do not tile the plane perfectly. This geometric limitation becomes critical at 80% concentration: the floes exhibit pronounced rebound behavior, which manifests either as sharp spikes in the DEM impulse — and thus a strong overestimation of resistance — or as a large number of floes being pushed laterally away from the ship, reducing contact and potentially underestimating the resistance.

Second, the absence of an ice channel in the simulation may affect the rebound dynamics between ice floes. In the experimental setup, an ice channel confines the domain laterally, meaning that floes pushed sideways by the ship are forced back by the channel walls. In the simulation, however, no such boundary condition exists. As a result, floes near the lateral boundaries ($y = \pm y_{max}$) are displaced outward through floe-floe interactions without any restoring effect, which likely leads to an overestimation of the resistance at high concentration (see Figure 25b).

Third, initial overlap between ice floes is observed in the experimental ice floe field, where some floes are stacked on top of others, as visible in Figure 25. This may alter the overall behavior of the ice field, and in particular the effective ice concentration seen by the ship.

These hypotheses could be investigated in future work to either confirm or rule out their respective contributions to the observed discrepancies at high ice concentration.



(a) Experiment **(b)** Simulation

Figure 25: Ice floe fields

4.3 Impact on the ship

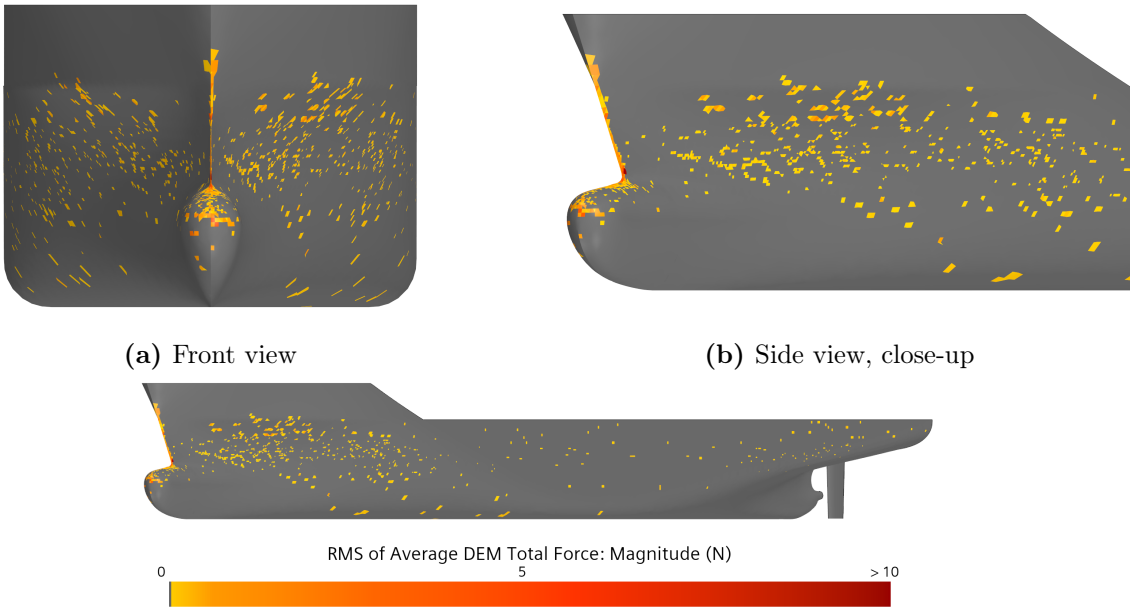
The regions of impact can be visualised in STAR-CCM+. In Figure 26, the root-mean-square (RMS) of the DEM contact force is plotted over the hull surface. The RMS provides a representative measure of the sustained loading at each location, giving more weight to the strongest impacts. It is defined as

$$F_{\mathrm{RMS}} = \sqrt{\frac{1}{T_{\mathrm{avg}}} \int_0^{T_{\mathrm{avg}}} F_{\mathrm{DEM}}^2(t)\,\mathrm{d}t}, \tag{51}$$

where $F_{\mathrm{DEM}}(t)$ is the DEM force acting on a given region of the hull and T is the averaging period.

The main impact zone is located at the edge of the bow, i.e. the most forward part of the hull at the free surface, where the ship first meets the oncoming floes. The bulbous bow is also strongly loaded. Finally, the sides of the hull experience some impacts as well, even near the keel; these originate from ice floes that have been submerged and pass beneath the hull before resurfacing.

Overall, the principal impacts occur on the frontal area of the hull. Beyond the estimation of the added resistance, the simulation therefore also allows the additional structural loads acting on the hull to be estimated. This is essential information for the design of a vessel intended to operate in Arctic waters, where these ice-induced loads can drive premature wear or structural damage. (see Figure 2).



(a) Front view

(b) Side view, close-up

(c) Side view, full hull

Figure 26: Distribution of the ice-floe impacts on the hull.

5 Additional work: particle clumps model investigation.

A preliminary study was conducted on the particle clumps model in STAR-CCM+. The objective of this study is to explore the capabilities offered by this model, and the issues it could address in the study of ships navigating in icy conditions.

The particle clumps model allows multiple sub-DEM particles in the form of spheres to represent a single ice floe. As a result, for the fluid-particle interaction, each sub-particle has its own interaction with the fluid.

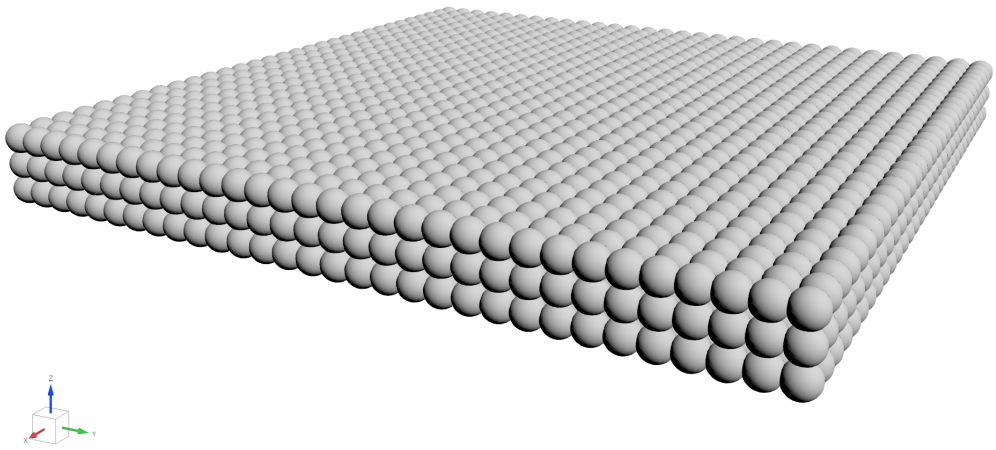
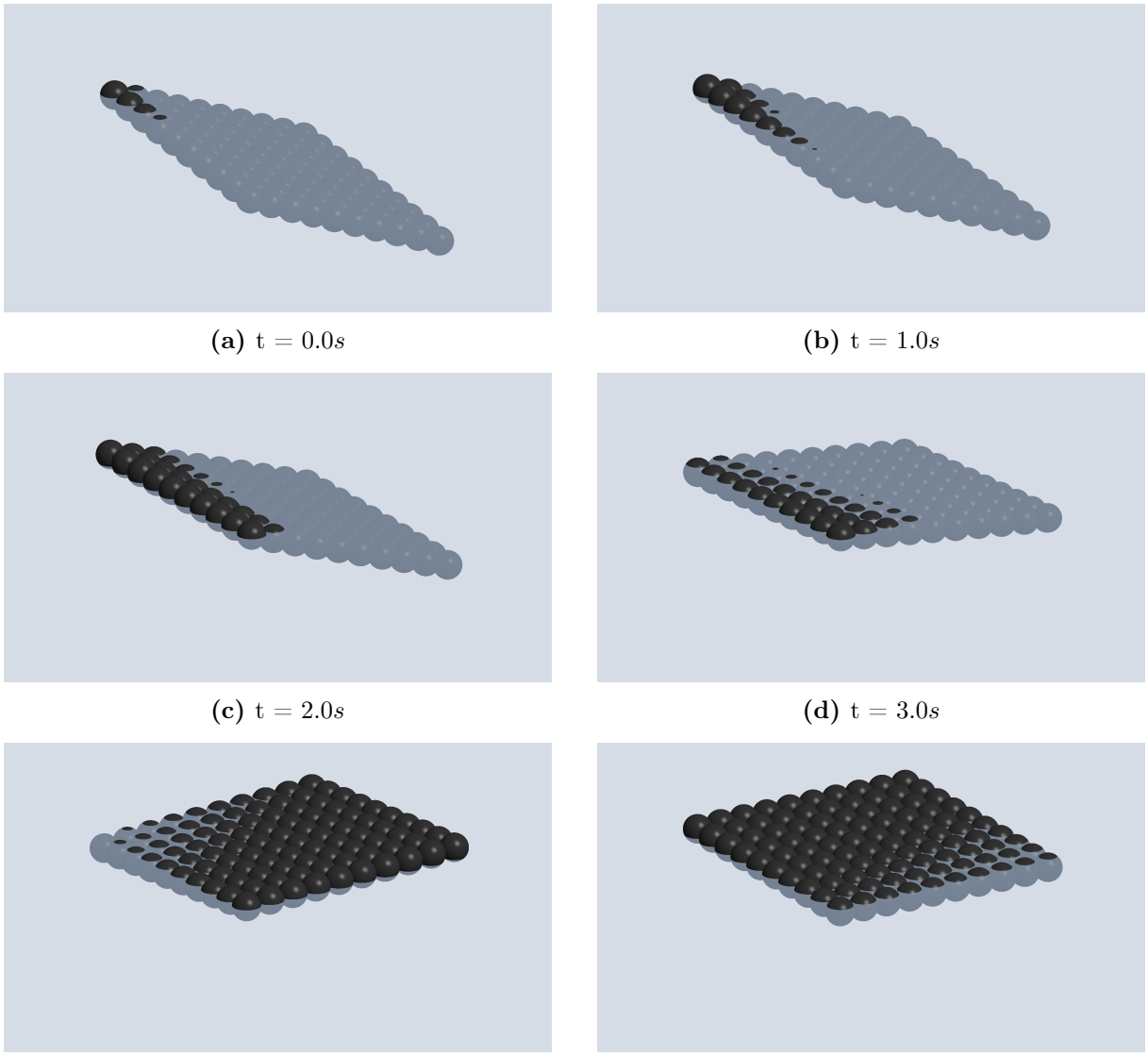


Figure 27: Example of an ice floe using the clumps model, size = $0.2 \times 0.2 \times 0.02$ m

This model therefore allows a more realistic representation of the fluid-particle interaction, as the surface forces are estimated from the individual sub-DEM particles. Consequently, if an ice floe is injected with an angle (about the x or y axis), it will return to its equilibrium position.



(a) t = 0.0s **(b)** t = 1.0s

(c) t = 2.0s **(d)** t = 3.0s

(e) t = 4.0s **(f)** t = 4.5s

Figure 28: Evolution of the ice floe position over time to the equilibrium, using particle clumps model

6 Conclusion

A CFD–DEM model was developed in STAR-CCM+ to predict the added resistance of a ship navigating through floating ice floes and validated on the KCS hull against the experimental results for ice concentrations of 60 %, 70 % and 80 % at five velocities ($F_r = 0.06$–0.18).

For 60% and 70%, the model tends to underestimate the total resistance. Numerical results are closer to experimental results at higher speeds (above 0.8 m/s), with relative errors near 10% or below, but they degrade at low speed (0.4 and 0.6 m/s), where relative errors on the total resistance reach 20 to 60%. This is attributed to the simplified hydrodynamic representation of the floes: the hydrostatic restoring moment that keeps a floe horizontal is absent, so the slightest contact makes a floe rotate prematurely. This interrupts the chain reaction through which upstream floes transmit their thrust to those in contact with the hull, leading to the underestimation. The floes should mostly slide along the hull, whereas in the simulations they rotate and submerge too often.

For $c_{\text{ice}} = 80\%$, the errors between simulated and experimental resistance are larger than at lower concentrations. There is still the floe-rotation problem identified at 60% and 70%. In addition, two supplementary factors could contribute as well: first, the shape of the ice floes could prove critical at high concentrations: circular floes allow lower maximum surface coverage than square ones for the same size distribution, which affects the contact behavior (for example, more inter-floe contacts for circular than square ones). At 80%, the circular floes are nearly in permanent contact; the resistance imparted by the ice floes on the ship no longer acts individually, but rather as a chain reaction with several floes contributing simultaneously. Secondly, the absence of an ice channel could be critical for comparison with the experimental data.

Despite these discrepancies, a resonance-like behavior consistent with experimental observations is nonetheless captured at $V = 0.6$ m/s, as discussed above.

Additionally, the moment of inertia scaling was modified to account for the added mass moment, but the value of this scaling depends strongly on the diameter of the ice floes. However, it was not possible to implement a per-floe moment of inertia scaling, so an average value was used instead.

Therefore, the current model allows one to evaluate, with relative reliability, the resistance to motion of a ship navigating in ice floes at high speeds and for ice concentrations of 60 % and 70 %, but shows its limitations when it comes to lower velocities and high concentrations.

7 Future Work

The prediction of ship resistance in ice remains an open research area with several promising directions.

First, regarding the limitations identified in this study, the choice of particle model is a likely source of error. Each ice floe is currently represented as a cylinder modelled by a single DEM particle, which prevents any hydrodynamic interaction with the surrounding fluid at the floe scale. One improvement would be to adopt a particle-clumps model, where each floe is composed of several sub-elements, enabling a more faithful representation of floe hydrodynamics. An initial study has been made (see Appendix 5)

Furthermore, cylindrical floe shapes were used on the basis that shape has been reported to have little influence on resistance [16]; however, the present results suggest that this conclusion may not hold at high ice concentrations. It could therefore be relevant to investigate square floes and, more importantly, realistic broken-ice shapes (random polygons), which would also allow concentrations approaching 100% to be simulated. A similar study has already been conducted using FEM [10], and could be extended to STAR-CCM+ using DEM with particle clumps, for which a preliminary study has been carried out with promising results.

A further avenue would be to investigate the CFD–DEM coupling scheme in greater depth. Several works consider *one-way* coupling sufficient for resistance prediction [16, 17], and direct comparisons show no notable difference in mean resistance between *one-way* and *two-way* schemes [17]. However, this conclusion has only been established at moderate concentrations: Huang et al. [16] justify the use of *one-way* coupling on the grounds that the concentration is not high enough to produce significant wave reflection, and limit their study to a maximum of 70%. At very high concentrations, where floes are densely packed and force chains can form between them, the feedback of the ice motion on the flow field has not been evaluated. It would therefore be worthwhile to assess whether *two-way* coupling becomes necessary in this regime.

Moreover, studying a ship in self-propulsion within ice floes could prove interesting. Indeed, in real-world scenarios, vessels cannot maintain a constant speed within an ice-floe field. It would also be interesting to conduct manoeuvrability tests.

Finally, the study of resistance in a brash-ice channel represents another interesting direction. The aspect ratio of brash-ice fragments is much closer to unity, meaning that single-particle DEM would already provide a more realistic representation without requiring particle clumps. Beyond the larger number of particles involved, the principal interest of this configuration lies in the boundary effects introduced by the ice channel walls.

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A Code

A.1 Ice Floe field random placement algorithm.

See Section 3.5.3. For this first algorithm, ice floes are randomly placed in decreasing order of diameter. It is fast but cannot reach high ice concentration ($c_{ice} = 60\%$ or less).

For example, the ice floe field generated for the KCS ($40s$ at maximum velocity $1.2m/s$) took 26 min.

```python
import pandas as pd
import matplotlib.pyplot as plt
import numpy as np
from matplotlib.patches import Circle, Rectangle
from pathlib import Path
import math
import os
import time


# ============================================================
# PARAMETERS  - edit this section only
# ============================================================

#  Physics
rho_water = 997.561
rho_ice   = 900
g         = 9.81

#  Ship
Lpp = 4.367

#  Injection geometry
x_injection = 7.0    # x-position of the injection plane [m]
y_injection = 5.0    # domain half-width in y [m]

#  Velocities to generate injection CSVs for
velocities = np.array([0.2, 0.4, 0.6, 0.8, 1.0, 1.2])   # [m/s]

#  Timing
time_step         = 0.01   # CFD time step [s]
time_first_impact = 40.0   # time when ice first reaches the ship bow [s]
duration_impact   = 40.0   # duration of the ice injection phase [s]

#  Floe geometry
thickness         = 0.02   # ice thickness [m]
ice_concentration = 0.60   # target ice concentration (0-1)

#  Size distribution - truncated log-normal on diameters
min_diameter   = 0.05    # [m]
max_diameter   = 0.40    # [m]
```

```
mu_diameter    = -1.59   # mean of ln(diameter)
sigma_diameter =  0.45   # std  of ln(diameter)

#  Output folder
OUTPUT_DIR = Path(__file__).resolve().parent /
↪  rf"output_IF_random_placement/C{int(ice_concentration * 100)}"
OUTPUT_DIR.mkdir(parents=True, exist_ok=True)


t_start = time.perf_counter()

# ============================================================
# BUILD ONE LARGE ICE FIELD FOR THE MAXIMUM VELOCITY
# ============================================================

max_velocity = np.max(velocities)

# The ice field must cover for {duration_impact} s at the maximum velocity
x_output = -duration_impact * max_velocity + x_injection

coords = np.array([
    [x_output, -y_injection],
    [x_injection, y_injection]
])

print("=" * 60)
print("Large ice field domain")
print("=" * 60)
print(f"Velocities:              {velocities}")
print(f"Maximum velocity:        {max_velocity:.3f} m/s")
print(f"Injection duration:      {duration_impact:.2f} s")
print(f"x_output:                {x_output:.3f} m")
print(f"coords =                 {coords}")
print()


# ============================================================
# GENERATE ICE FLOES
# ============================================================

area_domain = (coords[1][0] - coords[0][0]) * (coords[1][1] - coords[0][1])

ice_floes = []


def ice_floes_diameter_distribution(
    ice_floes,
    ice_concentration,
    area_domain,
```

```
    mu_diameter,
    sigma_diameter
):
    """Generate a distribution of ice floes that cover a given concentration of
    ↪  the surface."""
    total_ice_area = ice_concentration * area_domain

    current_ice_area = 0.0

    while current_ice_area < total_ice_area:

        diameter = np.random.lognormal(mu_diameter, sigma_diameter)

        if diameter > max_diameter or diameter < min_diameter:
            continue

        radius = diameter / 2
        area_floe = np.pi * radius ** 2

        ice_floes.append({
            "diameter": diameter,
            "area": area_floe
        })

        current_ice_area += area_floe

    return ice_floes, current_ice_area


ice_floes, current_ice_area = ice_floes_diameter_distribution(
    ice_floes,
    ice_concentration,
    area_domain,
    mu_diameter,
    sigma_diameter
)

print(
    f"Generated {len(ice_floes)} ice floes with a total area of "
    f"{current_ice_area:.2f} m², covering {current_ice_area / area_domain:.2%} "
    f"of the surface."
)

ice_floes = sorted(ice_floes, key=lambda x: x["diameter"], reverse=True)


# ============================================================
# PLACE ICE FLOES WITHOUT OVERLAP
# ============================================================
```

```

placed_floes = []

for i, floe in enumerate(ice_floes):

    diameter = floe["diameter"]
    print(f"Diameter of floe {i}: {diameter:.2f} m")

    radius = diameter / 2
    placed = False

    while placed is False:

        x = np.random.uniform(coords[0][0] + radius, coords[1][0] - radius)
        y = np.random.uniform(coords[0][1] + radius, coords[1][1] - radius)

        overlap = False

        for placed_floe in placed_floes:

            dx = x - placed_floe["x"]
            dy = y - placed_floe["y"]
            distance = np.sqrt(dx**2 + dy**2)

            if distance < (radius + placed_floe["radius"]):
                overlap = True
                break

            elif (
                (x - radius < coords[0][0])
                or (x + radius > coords[1][0])
                or (y - radius < coords[0][1])
                or (y + radius > coords[1][1])
            ):
                overlap = True
                break

        if not overlap:
            placed_floes.append({
                "x": x,
                "y": y,
                "radius": radius
            })
            placed = True
            break

    if not placed:
        print(f"Could not place floe with diameter {diameter:.2f} m.")

```

```
187
188 # ============================================================
189 # SAVE RAW ICE FLOE FIELD
190 # ============================================================
191
192 raw_field_df = pd.DataFrame([
193     {"x": f["x"], "y": f["y"], "radius": f["radius"]}
194     for f in placed_floes
195 ])
196 raw_field_path = OUTPUT_DIR / "ice_floe_field_raw.csv"
197 raw_field_df.to_csv(raw_field_path, index=False)
198 print(f"\nRaw ice floe field saved to: {raw_field_path}  ({len(placed_floes)}
    ↪  floes)")
199
200
201 # ============================================================
202 # PLOTTING
203 # ============================================================
204
205 def _plot_field(placed_floes, coords, title=None, filepath=None, show=True):
206     xmin, ymin = coords[0]
207     xmax, ymax = coords[1]
208     aspect = (xmax - xmin) / (ymax - ymin)
209     fig, ax = plt.subplots(figsize=(min(24, aspect * 4), 4))
210
211     for floe in placed_floes:
212         ax.add_patch(Circle(
213             (floe["x"], floe["y"]), floe["radius"],
214             edgecolor="steelblue", facecolor="lightblue",
215             linewidth=0.25, alpha=0.7
216         ))
217
218     ax.add_patch(Rectangle(
219         (xmin, ymin), xmax - xmin, ymax - ymin,
220         fill=False, edgecolor="red", linewidth=2
221     ))
222
223     ax.set_xlim(xmin - 0.1, xmax + 0.1)
224     ax.set_ylim(ymin - 0.1, ymax + 0.1)
225     ax.set_aspect("equal", adjustable="box")
226     ax.set_xlabel("x [m]")
227     ax.set_ylabel("y [m]")
228     ax.set_title(title or f"Ice floe field - {len(placed_floes)} floes")
229     ax.grid(True, linestyle="--", alpha=0.4)
230     plt.tight_layout()
231
232     if filepath:
233         fig.savefig(filepath, dpi=100, bbox_inches="tight")
234         print(f"  Saved: {filepath}")
```

```
    if show:
        plt.show()
    else:
        plt.close(fig)


def _plot_histogram(placed_floes, mu, sigma, min_d, max_d, filepath=None):
    diameters = [f["radius"] * 2 for f in placed_floes]
    d_range   = np.linspace(min_d, max_d, 300)
    d_trunc   = np.linspace(min_d, max_d, 3000)

    def lognorm_pdf(x):
        return (1.0 / (x * sigma * np.sqrt(2 * np.pi))
                * np.exp(-((np.log(x) - mu) ** 2) / (2 * sigma ** 2)))

    norm     = np.trapz(lognorm_pdf(d_trunc), d_trunc)
    pdf_norm = np.where(
        (d_range >= min_d) & (d_range <= max_d),
        lognorm_pdf(d_range) / norm, 0.0
    )

    fig, ax = plt.subplots(figsize=(8, 4))
    ax.hist(diameters, bins=30, density=True, alpha=0.6,
            edgecolor="black", label="Placed floes")
    ax.plot(d_range, pdf_norm, linewidth=2, label="Truncated log-normal PDF")
    ax.axvline(min_d, linestyle="--", linewidth=1.2)
    ax.axvline(max_d, linestyle="--", linewidth=1.2, label="Min / Max diameter")
    ax.set_xlabel("Diameter [m]")
    ax.set_ylabel("Probability density")
    ax.legend()
    ax.grid(True, linestyle="--", alpha=0.4)
    plt.tight_layout()

    if filepath:
        fig.savefig(filepath, dpi=150, bbox_inches="tight")
        print(f"  Saved: {filepath}")
    plt.show()


c_tag = f"C{int(ice_concentration * 100)}"

_plot_field(
    placed_floes, coords,
    title    = f"Ice floe field - C = {ice_concentration:.0%}
    ↪  ({len(placed_floes)} floes)",
    filepath = OUTPUT_DIR / f"ice_floe_field_{c_tag}.png",
    show     = False,
)

```

```python
_plot_histogram(
    placed_floes, mu_diameter, sigma_diameter,
    min_diameter, max_diameter,
    filepath = OUTPUT_DIR / f"diameter_histogram_{c_tag}.png",
)


# ============================================================
# FUNCTION TO EXPORT ONE CSV PER VELOCITY
# ============================================================

def save_injection_csv_for_velocity(
    velocity,
    placed_floes,
    coords,
    ice_concentration,
    thickness,
    duration_impact,
    time_first_impact,
    time_step,
    x_injection,
    Lpp,
    g,
    output_dir,
):
    """
    Export one injection CSV for a given velocity.

    The large ice field has already been generated.
    For each velocity, only the portion corresponding to duration_impact is
    ↪ used.
    """

    print()
    print(f"=== Export CSV for velocity = {velocity:.3f} m/s ===")

    start_injection_time = time_first_impact - (x_injection - Lpp) / velocity
    injection_end_time = start_injection_time + duration_impact

    Fr = np.sqrt(velocity**2 / (g * Lpp))

    xmin, ymin = coords[0]
    xmax, ymax = coords[1]

    domain_length = xmax - xmin

    extended_floes = sorted(placed_floes, key=lambda x: x["x"])

    print(f"start_injection_time = {start_injection_time:.3f} s")
```

```
    print(f"injection_end_time   = {injection_end_time:.3f} s")
    print(f"injection duration   = {duration_impact:.3f} s")
    print(f"Froude number        = {Fr:.4f}")
    print(f"Number of floes in large field: {len(extended_floes)}")

    t_bunch = 50 * time_step
    l_bunch = velocity * t_bunch

    print(f"t_bunch = {t_bunch:.6f} s")
    print(f"l_bunch = {l_bunch:.6f} m")

    injection_data_bunch = []

    current_time = start_injection_time + t_bunch

    z_equilibrium_cylinder = 0.0

    while current_time <= injection_end_time + 1e-12:

        x_min_bunch = (
            velocity * (current_time - start_injection_time - t_bunch)
            + xmin
        )

        x_max_bunch = (
            velocity * (current_time - start_injection_time)
            + xmin
        )

        for floe in extended_floes:

            if x_min_bunch < floe["x"] <= x_max_bunch:

                x_floe_adjusted = (
                    floe["x"]
                    - velocity * (current_time - start_injection_time)
                    + domain_length
                )

                injection_data_bunch.append({
                    "time": round(current_time - t_bunch, 10),
                    "x": x_floe_adjusted,
                    "y": floe["y"],
                    "z": z_equilibrium_cylinder,
                    "Radius": floe["radius"],
                    "thickness": thickness
                })

        current_time += t_bunch
```

```
    injection_bunch_df = pd.DataFrame(injection_data_bunch)

    velocity_str = f"{velocity:.2f}".replace(".", "p")
    Fr_str = f"{Fr:.3f}".replace(".", "p")

    filename = (
        f"C{int(ice_concentration * 100)}"
        f"_V{velocity_str}mps"
        f"_Fr{Fr_str}"
        f"_t{int(thickness * 1000)}mm"
        f"_tinject{start_injection_time:.2f}s".replace(".", "p")
        + f"_duration{int(duration_impact)}s"
        f".csv"
    )

    filepath = output_dir / filename

    injection_bunch_df.to_csv(filepath, index=False)

    print(f"Bunch injection data saved to: {filepath}")
    print(f"Number of injected floes: {len(injection_bunch_df)}")

    if len(injection_bunch_df) > 0:
        print(injection_bunch_df.head())
    else:
        print("Warning: no floes injected for this velocity.")

    return injection_bunch_df, filepath


# ============================================================
# EXPORT ONE CSV FOR EACH VELOCITY
# ============================================================

all_output_files = []

for velocity_i in velocities:

    injection_df, filepath = save_injection_csv_for_velocity(
        velocity=velocity_i,
        placed_floes=placed_floes,
        coords=coords,
        ice_concentration=ice_concentration,
        thickness=thickness,
        duration_impact=duration_impact,
        time_first_impact=time_first_impact,
        time_step=time_step,
        x_injection=x_injection,
```

```
        Lpp=Lpp,
        g=g,
        output_dir=OUTPUT_DIR,
    )

    all_output_files.append({
        "velocity_mps": velocity_i,
        "n_injected_floes": len(injection_df),
        "filepath": str(filepath)
    })


# ============================================================
# SAVE SUMMARY
# ============================================================

summary_df = pd.DataFrame(all_output_files)
summary_filepath = OUTPUT_DIR / "injection_files_summary.csv"
summary_df.to_csv(summary_filepath, index=False)

print()
print("All CSV files generated.")
print(f"Summary saved to: {summary_filepath}")
print(summary_df)

elapsed = time.perf_counter() - t_start
print(f"\nTotal run time : {int(elapsed // 60)} min {elapsed % 60:.1f} s")

plt.show()
```

A.2 Ice Floe field progressive growth algorithm.

See Section 3.5.3. This second algorithm is based on a progressive-growth strategy. This algorithm is really powerful to reach high ice concentration ($c_{ice} = 85\%$, maybe more), but take more time to generate.

For example, the ice floe field generated for the KCS ($40s$ at maximum velocity $1.2m/s$) took 79 min.

```python
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
from matplotlib.patches import Circle, Rectangle
from pathlib import Path
from scipy.spatial import cKDTree
import time


# ============================================================
# PARAMETERS  - edit this section only
# ============================================================


#  Physics
rho_water = 997.561
rho_ice   = 900.0
g         = 9.81

#  Ship
Lpp = 4.367

#  Injection geometry
x_injection = 10.0    # x-position of the injection plane [m]
y_injection = 2.5     # domain half-width in y [m]

#  Velocities to generate injection CSVs for
velocities = np.array([0.4, 0.6, 0.8, 1.0, 1.2])   # [m/s]

#  Timing
time_step         = 0.01   # CFD time step [s]
time_first_impact = 40.0   # time when ice first reaches the ship bow [s]
duration_impact   = 20.0   # duration of the ice injection phase [s]

#  Floe geometry
thickness         = 0.02   # ice thickness [m]
ice_concentration = 0.50   # target ice concentration (0-1)

#  Minimum gap between floes
# Guarantees that every pair of floes satisfies:
#   dist(i, j) >= R_i + R_j + min_gap
# The packing internally inflates each radius by min_gap/2 (effective radius).
```

```
# The exported CSV always contains the real (un-inflated) radii.
# Set to 0.0 to allow floes to touch (original behaviour).
min_gap = 0.003   # [m]  e.g. 2 mm

#  Size distribution - truncated log-normal on diameters
min_diameter   = 0.05    # [m]
max_diameter   = 0.40    # [m]
mu_diameter    = -1.59   # mean of ln(diameter)
sigma_diameter =  0.45   # std  of ln(diameter)

#  Reproducibility
random_seed = 17
rng = np.random.default_rng(random_seed)

#  Progressive-growth tuning
initial_radius_scale      = 0.40   # start floes at this fraction of effective
↪  radius
initial_growth_step       = 0.04
minimum_growth_step       = 0.001
growth_step_increase      = 1.15
growth_step_decrease      = 0.50

#  Relaxation
relaxation_steps_per_growth = 2000
final_relaxation_steps      = 12000
relaxation_factor           = 0.45
target_max_overlap          = 1e-6   # [m] - in effective-radius space

#  Initial placement
# Best-clearance strategy: test n_candidates random positions per floe
# and keep the one with maximum clearance from already-placed floes.
n_candidates_initial = 2000

#  Relocation (escape from stuck configurations)
enable_relocation     = True
relocation_fraction   = 0.10
relocation_candidates = 2000

#  Final shrink
enable_final_shrink    = True
tiny_overlap_threshold = 1e-6
final_shrink_safety    = 1e-12

#  Intermediate growth plots
save_growth_plots  = False
plot_every_n_steps = 10


BASE_DIR   = Path(__file__).resolve().parent / "output_IF_progressive_growth"
```

```python
OUTPUT_DIR = BASE_DIR / rf"C{int(ice_concentration * 100)}"
OUTPUT_DIR.mkdir(parents=True, exist_ok=True)
PLOT_DIR   = BASE_DIR / "plots_growth"
PLOT_DIR.mkdir(parents=True, exist_ok=True)


# ============================================================
# DOMAIN SIZING  (based on max velocity and injection duration)
# ============================================================

max_velocity = float(np.max(velocities))
x_output     = x_injection - duration_impact * max_velocity   # leftmost domain edge

coords = np.array([
    [x_output,    -y_injection],
    [x_injection,  y_injection],
])

# KDTree interaction radius: two effective discs can interact if
#   dist < (R_eff_i + R_eff_j)_max = max_effective_diameter
# where R_eff = R + min_gap/2    max_effective_diameter = max_diameter + min_gap
_MAX_INTERACT = max_diameter + min_gap

print("=" * 60)
print("Ice floe generator - progressive-growth packing")
print(f"  Target concentration : {ice_concentration:.0%}")
print(f"  Minimum gap          : {min_gap * 1000:.1f} mm")
print(f"  Velocities           : {velocities} m/s")
print(f"  Domain x             : [{x_output:.3f}, {x_injection}] m")
print(f"  Domain y             : [{-y_injection}, {y_injection}] m")
print(f"  Domain area          : {(x_injection - x_output) * 2 * y_injection:.3f} m²")
print("=" * 60)
print()


# ============================================================
# GEOMETRY HELPERS
# ============================================================

def domain_bounds(coords):
    (xmin, ymin), (xmax, ymax) = coords
    return xmin, ymin, xmax, ymax


def domain_area(coords):
    xmin, ymin, xmax, ymax = domain_bounds(coords)
    return (xmax - xmin) * (ymax - ymin)
```

```python


def achieved_concentration(radii, coords):
    """Ice concentration based on actual (non-inflated) radii."""
    if len(radii) == 0:
        return 0.0
    return float(np.sum(np.pi * np.asarray(radii) ** 2) / domain_area(coords))


def clamp_to_domain(positions, radii, coords):
    """Keep every floe centre at least one radius away from each wall."""
    xmin, ymin, xmax, ymax = domain_bounds(coords)
    positions[:, 0] = np.clip(positions[:, 0], xmin + radii, xmax - radii)
    positions[:, 1] = np.clip(positions[:, 1], ymin + radii, ymax - radii)
    return positions


# ============================================================
# OVERLAP HELPERS  (KDTree-accelerated)
# NOTE: pass *effective* radii (R + min_gap/2) to enforce the gap.
# ============================================================

def _overlapping_pairs(positions, eff_radii):
    """
    Return (ia, ja, overlaps) for all pairs where
        dist(i, j) < eff_R_i + eff_R_j
    i.e. where the effective-radius discs overlap.
    """
    if len(eff_radii) < 2:
        empty = np.array([], dtype=int)
        return empty, empty, np.array([])

    tree  = cKDTree(positions)
    pairs = tree.query_pairs(_MAX_INTERACT)
    if not pairs:
        empty = np.array([], dtype=int)
        return empty, empty, np.array([])

    ia = np.fromiter((p[0] for p in pairs), dtype=int, count=len(pairs))
    ja = np.fromiter((p[1] for p in pairs), dtype=int, count=len(pairs))
    dx   = positions[ia, 0] - positions[ja, 0]
    dy   = positions[ia, 1] - positions[ja, 1]
    dist = np.sqrt(dx**2 + dy**2)
    ov   = eff_radii[ia] + eff_radii[ja] - dist
    mask = ov > 0
    return ia[mask], ja[mask], ov[mask]


def overlap_statistics(positions, eff_radii):
```

```
    """Return (n_overlaps, max_overlap, penalty, score_per_floe)."""
    ia, ja, ov = _overlapping_pairs(positions, eff_radii)
    n_ov    = len(ov)
    max_ov  = float(ov.max()) if n_ov > 0 else 0.0
    penalty = float(np.sum(ov ** 2))
    score   = np.zeros(len(eff_radii))
    if n_ov > 0:
        np.add.at(score, ia, ov ** 2)
        np.add.at(score, ja, ov ** 2)
    return n_ov, max_ov, penalty, score


# ============================================================
# MIN-GAP VERIFICATION
# ============================================================

def check_min_gap(positions, actual_radii, min_gap, tol=1e-9):
    """
    Verify that every pair of floes satisfies:
        dist(i, j) >= R_i + R_j + min_gap  (up to tol)

    Returns
    -------
    n_violations : int      number of pairs below the required gap
    min_clearance : float   smallest edge-to-edge clearance found [m]
    """
    if min_gap <= 0.0 or len(actual_radii) < 2:
        return 0, np.inf

    tree  = cKDTree(positions)
    pairs = tree.query_pairs(_MAX_INTERACT)

    n_viol  = 0
    min_clr = np.inf
    for i, j in pairs:
        dx   = positions[i, 0] - positions[j, 0]
        dy   = positions[i, 1] - positions[j, 1]
        dist = np.sqrt(dx**2 + dy**2)
        clr  = dist - actual_radii[i] - actual_radii[j]
        if clr < min_clr:
            min_clr = clr
        if clr < min_gap - tol:
            n_viol += 1

    return n_viol, float(min_clr)


# ============================================================
# RELAXATION  (KDTree-accelerated, works on effective radii)
```

```
# ============================================================

def relax_overlaps(positions, eff_radii, coords, rng,
                   n_steps=1000, factor=0.45,
                   target_max_overlap=1e-6, verbose=False):
    """
    Soft-disc relaxation on effective radii.
    Overlapping pairs push each other apart proportionally to their
    inverse mass (smaller floes move more).
    """
    positions = positions.copy()
    eff_radii = np.asarray(eff_radii)
    eps       = 1e-12
    inv_mass  = 1.0 / (eff_radii ** 2 + eps)

    for step in range(n_steps):
        tree  = cKDTree(positions)
        pairs = tree.query_pairs(_MAX_INTERACT)
        if not pairs:
            break

        ia = np.fromiter((p[0] for p in pairs), dtype=int, count=len(pairs))
        ja = np.fromiter((p[1] for p in pairs), dtype=int, count=len(pairs))
        dx   = positions[ia, 0] - positions[ja, 0]
        dy   = positions[ia, 1] - positions[ja, 1]
        dist = np.sqrt(dx**2 + dy**2)
        ov   = eff_radii[ia] + eff_radii[ja] - dist
        mask = ov > 0
        n_ov   = int(mask.sum())
        max_ov = float(ov[mask].max()) if n_ov > 0 else 0.0

        if verbose and step % 200 == 0:
            pen = float(np.sum(ov[mask] ** 2)) if n_ov > 0 else 0.0
            print(f"    step {step:5d} | overlaps {n_ov:5d} | "
                  f"max_ov {max_ov:.3e} m | pen {pen:.3e}")

        if n_ov == 0 or max_ov <= target_max_overlap:
            break

        ia_m = ia[mask]; ja_m = ja[mask]
        ov_m = ov[mask]
        dx_m = dx[mask]; dy_m = dy[mask]
        dist_m = dist[mask]

        tiny        = dist_m < eps
        rand_angles = rng.uniform(0, 2 * np.pi, int(tiny.sum()))
        nx = np.where(tiny,
                      np.cos(rand_angles[np.cumsum(tiny) - 1]) if
                      ↪  rand_angles.size else 0.0,
```

```
                        dx_m / np.where(tiny, 1.0, dist_m))
        ny = np.where(tiny,
                        np.sin(rand_angles[np.cumsum(tiny) - 1]) if
                        ↪  rand_angles.size else 0.0,
                        dy_m / np.where(tiny, 1.0, dist_m))

        inv_sum = inv_mass[ia_m] + inv_mass[ja_m]
        si      = inv_mass[ia_m] / inv_sum
        sj      = inv_mass[ja_m] / inv_sum
        corr    = factor * ov_m

        disp_x = np.zeros(len(eff_radii))
        disp_y = np.zeros(len(eff_radii))
        np.add.at(disp_x, ia_m,  corr * si * nx)
        np.add.at(disp_y, ia_m,  corr * si * ny)
        np.add.at(disp_x, ja_m, -corr * sj * nx)
        np.add.at(disp_y, ja_m, -corr * sj * ny)

        positions[:, 0] += disp_x
        positions[:, 1] += disp_y
        positions = clamp_to_domain(positions, eff_radii, coords)

    return positions


# ============================================================
# STEP 1 - GENERATE RADII
# ============================================================

def generate_radii(coords, target_concentration, mu, sigma,
                   min_d, max_d, rng):
    """
    Draw diameters from a truncated log-normal distribution until the
    cumulative actual ice area >= target_concentration * domain_area.
    Returns actual (non-inflated) radii sorted largest-first.
    """
    target_area  = target_concentration * domain_area(coords)
    radii        = []
    current_area = 0.0

    while current_area < target_area:
        d = rng.lognormal(mu, sigma)
        if d < min_d or d > max_d:
            continue
        r = 0.5 * d
        radii.append(r)
        current_area += np.pi * r ** 2

    radii = np.sort(np.array(radii))[::-1]
```

```
332
333     print("Diameter generation")
334     print("-------------------")
335     print(f"  Floes generated          : {len(radii)}")
336     print(f"  Target concentration     : {target_concentration:.2%}")
337     print(f"  Achieved concentration   : {achieved_concentration(radii,
        ↪  coords):.2%}")
338     print(f"  Diameter range           : {2 * radii[-1]:.3f} - {2 * radii[0]:.3f}
        ↪  m")
339     print(f"  Mean diameter            : {2 * np.mean(radii):.3f} m")
340     print()
341     return radii
342
343
344 # ============================================================
345 # STEP 2 - INITIAL PLACEMENT  (best-clearance + KDTree)
346 # ============================================================
347
348 def place_initial_shrunk_floes(eff_final_radii, coords, rng,
349                                initial_scale=0.40,
350                                n_candidates=2000):
351     """
352     Place floes at `initial_scale × eff_final_radius` using a
353     best-clearance strategy (from IF_generator_algoVI_domain_star.py)
354     accelerated by a KDTree.
355
356     For each floe, `n_candidates` random positions are tested and the one
357     with maximum clearance from already-placed floes is kept.  If a
358     position with strictly positive clearance is found it is used
359     immediately without testing the remaining candidates.
360     """
361     xmin, ymin, xmax, ymax = domain_bounds(coords)
362     working_eff_radii = initial_scale * eff_final_radii
363     n                 = len(working_eff_radii)
364     positions         = np.zeros((n, 2))
365
366     tree          = None
367     tree_built_at = 0
368     rebuild_every = 100
369
370     for k, r in enumerate(working_eff_radii):
371
372         # Rebuild KDTree periodically
373         if k > 0 and (k - tree_built_at) >= rebuild_every:
374             tree = cKDTree(positions[:k])
375             tree_built_at = k
376
377         best_pos = np.array([rng.uniform(xmin + r, xmax - r),
378                              rng.uniform(ymin + r, ymax - r)])
```

```
        best_clr = -np.inf

        for _ in range(n_candidates):
            cand = np.array([rng.uniform(xmin + r, xmax - r),
                             rng.uniform(ymin + r, ymax - r)])

            if k == 0:
                best_pos = cand
                best_clr = np.inf
                break

            # Fast neighbour lookup
            if tree is not None:
                nearby = tree.query_ball_point(cand, _MAX_INTERACT *
                ↪  initial_scale + r)
            else:
                nearby = list(range(k))

            if not nearby:
                # No neighbours → perfect spot, stop immediately
                best_pos = cand
                best_clr = np.inf
                break

            pos_n = positions[np.array(nearby)]
            r_n   = working_eff_radii[np.array(nearby)]
            dists = np.sqrt(((pos_n[:, 0] - cand[0]) ** 2
                             + (pos_n[:, 1] - cand[1]) ** 2))
            clr   = float(np.min(dists - r_n - r))

            if clr > best_clr:
                best_clr = clr
                best_pos = cand

        positions[k] = best_pos

        if (k + 1) % 500 == 0 or k + 1 == n:
            c = achieved_concentration(working_eff_radii[:k + 1], coords)
            print(f"  Placement: {k+1:5d}/{n} floes | "
                  f"C_eff = {c:.1%} | best_clr = {best_clr:.3e} m")

    n_ov, max_ov, _, _ = overlap_statistics(positions, working_eff_radii)
    print()
    print("Initial placement complete")
    print(f"  Radius scale   : {initial_scale:.3f}")
    print(f"  C_eff          : {achieved_concentration(working_eff_radii,
    ↪  coords):.2%}")
    print(f"  Overlaps       : {n_ov}   max = {max_ov:.3e} m")
    print()
```

```
    return positions


# ============================================================
# STEP 3 - RELOCATION  (escape stuck configurations)
# ============================================================

def relocate_problematic_floes(positions, eff_radii, coords, rng,
                               fraction=0.10, n_candidates=2000):
    """
    Move the most-overlapping floes to better positions (best-clearance
    among n_candidates random spots, KDTree-accelerated).
    """
    positions = positions.copy()
    xmin, ymin, xmax, ymax = domain_bounds(coords)

    _, _, _, score = overlap_statistics(positions, eff_radii)
    n_move      = max(1, int(fraction * len(eff_radii)))
    problematic = np.where(score > 0)[0]
    if len(problematic) == 0:
        return positions

    ranking  = score[problematic] / (eff_radii[problematic] ** 2 + 1e-12)
    order    = np.argsort(ranking)[::-1]
    selected = problematic[order[:min(n_move, len(problematic))]]

    tree = cKDTree(positions)

    for i in selected:
        r = eff_radii[i]

        # Current clearance
        nearby_cur = tree.query_ball_point(positions[i], _MAX_INTERACT + r)
        if i in nearby_cur:
            nearby_cur.remove(i)
        if nearby_cur:
            pos_n = positions[np.array(nearby_cur)]
            r_n   = eff_radii[np.array(nearby_cur)]
            dists = np.sqrt(((pos_n[:, 0] - positions[i, 0]) ** 2
                             + (pos_n[:, 1] - positions[i, 1]) ** 2))
            best_clr = float(np.min(dists - r_n - r))
        else:
            best_clr = np.inf
        best_pos = positions[i].copy()

        for _ in range(n_candidates):
            cand   = np.array([rng.uniform(xmin + r, xmax - r),
                               rng.uniform(ymin + r, ymax - r)])
            nearby = tree.query_ball_point(cand, _MAX_INTERACT + r)
```

```
            if i in nearby:
                nearby.remove(i)
            if not nearby:
                best_pos = cand
                break
            pos_n = positions[np.array(nearby)]
            r_n   = eff_radii[np.array(nearby)]
            dists = np.sqrt(((pos_n[:, 0] - cand[0]) ** 2
                             + (pos_n[:, 1] - cand[1]) ** 2))
            clr   = float(np.min(dists - r_n - r))
            if clr > best_clr:
                best_clr = clr
                best_pos = cand

        positions[i] = best_pos

    return positions


# ============================================================
# STEP 4 - PROGRESSIVE GROWTH MAIN LOOP
# ============================================================

def progressive_growth_packing(
    eff_final_radii, coords, rng,
    initial_scale=0.40, initial_growth_step=0.04,
    minimum_growth_step=0.001, growth_step_increase=1.15,
    growth_step_decrease=0.50, relax_steps=2000,
    final_relax_steps=12000, relax_factor=0.45,
    target_max_overlap=1e-6, enable_relocation=True,
    reloc_fraction=0.10, reloc_candidates=2000,
    n_cand_initial=2000, save_plots=False, plot_every=10,
):
    """
    Progressive-growth packing on effective radii.

    Grows floes from `initial_scale * eff_final_radii` to `eff_final_radii`
    in adaptive steps, relaxing overlaps after each step.  Returns
    (positions, eff_final_radii_sorted).
    """
    eff_final_radii = np.sort(np.asarray(eff_final_radii))[::-1]

    positions = place_initial_shrunk_floes(
        eff_final_radii, coords, rng,
        initial_scale=initial_scale,
        n_candidates=n_cand_initial,
    )

    scale         = initial_scale
```

```
    growth_step   = initial_growth_step
    current_eff_r = scale * eff_final_radii
    growth_id     = 0

    # Initial relaxation
    positions = relax_overlaps(
        positions, current_eff_r, coords, rng,
        n_steps=relax_steps, factor=relax_factor,
        target_max_overlap=target_max_overlap,
    )

    while scale < 1.0 - 1e-12:
        trial_scale = min(1.0, scale + growth_step)
        trial_eff_r = trial_scale * eff_final_radii

        print(f"\nGrowth {growth_id:4d} | "
              f"scale {scale:.4f} -> {trial_scale:.4f} | "
              f"C_eff = {achieved_concentration(trial_eff_r, coords):.2%} | "
              f"step = {growth_step:.4f}")

        trial_pos = relax_overlaps(
            positions.copy(), trial_eff_r, coords, rng,
            n_steps=relax_steps, factor=relax_factor,
            target_max_overlap=target_max_overlap,
        )

        n_ov, max_ov, pen, _ = overlap_statistics(trial_pos, trial_eff_r)
        print(f"  After relax  | overlaps {n_ov:5d} | "
              f"max_ov {max_ov:.3e} m | pen {pen:.3e}")

        success = (n_ov == 0 or max_ov <= target_max_overlap)

        if not success and enable_relocation:
            print("  Relocating stuck floes...")
            trial_pos = relocate_problematic_floes(
                trial_pos, trial_eff_r, coords, rng,
                fraction=reloc_fraction, n_candidates=reloc_candidates,
            )
            trial_pos = relax_overlaps(
                trial_pos, trial_eff_r, coords, rng,
                n_steps=relax_steps, factor=relax_factor,
                target_max_overlap=target_max_overlap,
            )
            n_ov, max_ov, pen, _ = overlap_statistics(trial_pos, trial_eff_r)
            print(f"  After reloc  | overlaps {n_ov:5d} | "
                  f"max_ov {max_ov:.3e} m | pen {pen:.3e}")
            success = (n_ov == 0 or max_ov <= target_max_overlap)

        if success:
```

```
            positions     = trial_pos
            scale         = trial_scale
            current_eff_r = trial_eff_r
            growth_step   = min(growth_step * growth_step_increase, 0.05)
            print("  Growth accepted.")

            if save_plots and growth_id % plot_every == 0:
                _plot_field(
                    positions, current_eff_r, coords,
                    title=(f"Growth {growth_id} - scale {scale:.3f} - "
                           f"C_eff = {achieved_concentration(current_eff_r,
                           ↪  coords):.1%}"),
                    filepath=PLOT_DIR / f"growth_{growth_id:04d}.png",
                    show=False,
                )
        else:
            growth_step *= growth_step_decrease
            print(f"  Growth rejected. New step = {growth_step:.4f}")
            if growth_step < minimum_growth_step:
                print("  Minimum growth step reached - accepting and exiting.")
                positions     = trial_pos
                scale         = trial_scale
                current_eff_r = trial_eff_r
                break

        growth_id += 1

    #  Final relaxation
    print()
    print("Final relaxation")
    print("----------------")
    positions = relax_overlaps(
        positions, eff_final_radii, coords, rng,
        n_steps=final_relax_steps, factor=relax_factor,
        target_max_overlap=target_max_overlap, verbose=True,
    )

    #  Final relocation passes
    if enable_relocation:
        _, max_ov, _, _ = overlap_statistics(positions, eff_final_radii)
        if max_ov > target_max_overlap:
            print()
            print("Final relocation passes")
            print("-----------------------")
            for pass_id in range(5):
                positions = relocate_problematic_floes(
                    positions, eff_final_radii, coords, rng,
                    fraction=reloc_fraction, n_candidates=reloc_candidates,
                )
```

```
                positions = relax_overlaps(
                    positions, eff_final_radii, coords, rng,
                    n_steps=final_relax_steps // 2, factor=relax_factor,
                    target_max_overlap=target_max_overlap,
                )
                n_ov, max_ov, pen, _ = overlap_statistics(positions,
                ↪  eff_final_radii)
                print(f"  Pass {pass_id + 1} | overlaps {n_ov} | "
                      f"max_ov {max_ov:.3e} m | pen {pen:.3e}")
                if n_ov == 0 or max_ov <= target_max_overlap:
                    break

    return positions, eff_final_radii


# ============================================================
# STEP 5 - FINAL TINY-OVERLAP SHRINK
# ============================================================

def final_shrink(positions, eff_radii,
                 threshold=1e-6, safety=1e-12, enable=True):
    """
    If the maximum effective-radius overlap is smaller than `threshold`,
    shrink every effective radius by (max_overlap / 2 + safety) to
    guarantee zero overlap.  The concentration check uses effective radii
    so the condition is satisfied whenever actual concentration >= target.
    """
    eff_radii = np.asarray(eff_radii).copy()
    n_ov, max_ov, pen, _ = overlap_statistics(positions, eff_radii)

    print()
    print("Final shrink check")
    print("------------------")
    print(f"  Overlaps   : {n_ov}   max = {max_ov:.3e} m")

    if not enable:
        print("  Disabled.")
        return eff_radii, 0.0, n_ov, max_ov, pen

    if n_ov > 0 and max_ov < threshold:
        shrink    = 0.5 * max_ov + safety
        eff_radii -= shrink
        n_ov2, max_ov2, pen2, _ = overlap_statistics(positions, eff_radii)
        print(f"  Shrink applied : {shrink:.3e} m")
        print(f"  Overlaps left  : {n_ov2}")
        return eff_radii, shrink, n_ov2, max_ov2, pen2

    print("  No shrink needed." if n_ov == 0 else
          "  Max overlap >= threshold - no shrink applied.")
```

```
    return eff_radii, 0.0, n_ov, max_ov, pen


# ============================================================
# PLOTTING
# ============================================================

def _plot_field(positions, radii, coords, title=None, filepath=None, show=True):
    xmin, ymin, xmax, ymax = domain_bounds(coords)
    aspect = (xmax - xmin) / (ymax - ymin)
    fig, ax = plt.subplots(figsize=(min(24, aspect * 4), 4))

    for (px, py), r in zip(positions, radii):
        ax.add_patch(Circle((px, py), r,
                            edgecolor="steelblue", facecolor="lightblue",
                            linewidth=0.25, alpha=0.7))

    ax.add_patch(Rectangle((xmin, ymin), xmax - xmin, ymax - ymin,
                            fill=False, edgecolor="red", linewidth=2))

    c = achieved_concentration(radii, coords)
    ax.set_xlim(xmin - 0.5, xmax + 0.5)
    ax.set_ylim(ymin - 0.3, ymax + 0.3)
    ax.set_aspect("equal")
    ax.set_xlabel("x [m]")
    ax.set_ylabel("y [m]")
    ax.set_title(title or f"Ice floe field - C = {c:.1%}  ({len(radii)} floes)")
    ax.grid(True, linestyle="--", alpha=0.4)
    plt.tight_layout()

    if filepath:
        fig.savefig(filepath, dpi=100, bbox_inches="tight")
        print(f"  Saved: {filepath}")
    if show:
        plt.show()
    else:
        plt.close(fig)


def _plot_histogram(radii, mu, sigma, min_d, max_d, filepath=None):
    diameters = 2.0 * np.asarray(radii)
    d_range   = np.linspace(min_d, max_d, 300)
    d_trunc   = np.linspace(min_d, max_d, 3000)

    def lognorm_pdf(x):
        return (1.0 / (x * sigma * np.sqrt(2 * np.pi))
                * np.exp(-((np.log(x) - mu) ** 2) / (2 * sigma ** 2)))

    norm     = np.trapz(lognorm_pdf(d_trunc), d_trunc)
```

```
    pdf_norm = np.where((d_range >= min_d) & (d_range <= max_d),
                        lognorm_pdf(d_range) / norm, 0.0)

    fig, ax = plt.subplots(figsize=(8, 4))
    ax.hist(diameters, bins=30, density=True, alpha=0.6,
            edgecolor="black", label="Placed floes")
    ax.plot(d_range, pdf_norm, linewidth=2, label="Truncated log-normal PDF")
    ax.axvline(min_d, linestyle="--", linewidth=1.2)
    ax.axvline(max_d, linestyle="--", linewidth=1.2, label="Min / Max diameter")
    ax.set_xlabel("Diameter [m]")
    ax.set_ylabel("Probability density")
    ax.legend()
    ax.grid(True, linestyle="--", alpha=0.4)
    plt.tight_layout()

    if filepath:
        fig.savefig(filepath, dpi=150, bbox_inches="tight")
        print(f"  Saved: {filepath}")
    plt.show()


# ============================================================
# INJECTION CSV EXPORT  (one per velocity)
# ============================================================

def save_injection_csv(velocity, placed_floes, coords,
                       ice_concentration, min_gap, thickness,
                       duration_impact, time_first_impact,
                       time_step, x_injection, Lpp,
                       g, output_dir):
    """
    Write the injection CSV for one velocity.

    The field is divided into time-bunches of width t_bunch = 50 * dt.
    At each bunch time, floes whose reference x-coordinate falls within
    the current bundle window are emitted at the inlet x_injection.
    """
    print(f"\n  velocity = {velocity:.3f} m/s")

    Fr      = velocity / np.sqrt(g * Lpp)
    t_start = time_first_impact - (x_injection - Lpp) / velocity
    t_end   = t_start + duration_impact
    t_bunch = 50 * time_step
    xmin    = coords[0, 0]
    dom_len = coords[1, 0] - xmin

    sorted_floes = sorted(placed_floes, key=lambda f: f["x"])

    print(f"    Froude number    : {Fr:.4f}")
```

```
    print(f"    t_inject start   : {t_start:.3f} s")
    print(f"    t_inject end     : {t_end:.3f} s")
    print(f"    Floes in field   : {len(sorted_floes)}")

    rows         = []
    current_time = t_start + t_bunch

    while current_time <= t_end + 1e-12:
        x_win_lo = velocity * (current_time - t_start - t_bunch) + xmin
        x_win_hi = velocity * (current_time - t_start)           + xmin

        for floe in sorted_floes:
            if x_win_lo < floe["x"] <= x_win_hi:
                x_adj = floe["x"] - velocity * (current_time - t_start) +
                ↪  dom_len
                rows.append({
                    "time"     : round(current_time - t_bunch, 10),
                    "x"        : x_adj,
                    "y"        : floe["y"],
                    "z"        : 0.0,
                    "Radius"   : floe["radius"],   # actual (non-inflated)
                    ↪  radius
                    "thickness": thickness,
                })

        current_time += t_bunch

    df = pd.DataFrame(rows)

    v_str   = f"{velocity:.2f}".replace(".", "p")
    Fr_str  = f"{Fr:.3f}".replace(".", "p")
    t_str   = f"{t_start:.2f}s".replace(".", "p")
    gap_str = f"{int(round(min_gap * 1000))}mm"
    fname   = (f"C{int(ice_concentration * 100)}"
               f"_V{v_str}mps"
               f"_Fr{Fr_str}"
               f"_t{int(thickness * 1000)}mm"
               f"_gap{gap_str}"
               f"_tinject{t_str}"
               f"_duration{int(duration_impact)}s.csv")

    fp = output_dir / fname
    df.to_csv(fp, index=False)
    print(f"    Saved ({len(df)} rows): {fp.name}")
    return df, fp


# ============================================================
# SNAPSHOT INJECTION CSV  (whole field at once - visual check)
```

```
# ============================================================

def save_snapshot_injection_csv(placed_floes, ice_concentration, min_gap,
                                thickness, time_first_impact, output_dir):
    """
    Dump the entire ice-floe field in a single injection CSV.

    All floes share the same time stamp (time_first_impact) and keep their
    packed (x, y) coordinates exactly.  Injecting this file in STAR-CCM+
    places every floe simultaneously at t = time_first_impact, which lets
    you verify the field layout (concentration, size distribution, gaps)
    before committing to the full time-bunched injection.

    The file is intentionally velocity-independent: it is purely a spatial
    snapshot of the packed field.
    """
    rows = [
        {
            "time"     : time_first_impact,
            "x"        : floe["x"],
            "y"        : floe["y"],
            "z"        : 0.0,
            "Radius"   : floe["radius"],
            "thickness": thickness,
        }
        for floe in placed_floes
    ]

    df      = pd.DataFrame(rows)
    gap_str = f"{int(round(min_gap * 1000))}mm"
    fname   = (f"C{int(ice_concentration * 100)}"
               f"_gap{gap_str}"
               f"_SNAPSHOT_t{time_first_impact:.2f}s.csv")

    fp = output_dir / fname
    df.to_csv(fp, index=False)
    print(f"  Snapshot CSV saved ({len(df)} floes at
    ↪  t={time_first_impact:.2f}s): {fp.name}")
    return df, fp


# ============================================================
# MAIN
# ============================================================

if __name__ == "__main__":

    t_start = time.perf_counter()

```

```
    # ------------------------------------------------------------------
    # 1.  Generate actual radii to reach target concentration
    # ------------------------------------------------------------------
    final_radii = generate_radii(
        coords, ice_concentration,
        mu_diameter, sigma_diameter,
        min_diameter, max_diameter, rng,
    )

    # ------------------------------------------------------------------
    # 2.  Build effective radii  R_eff = R + min_gap/2
    #     Packing on R_eff guarantees dist(i,j) >= R_i + R_j + min_gap.
    # ------------------------------------------------------------------
    eff_final_radii = final_radii + 0.5 * min_gap

    c_actual = achieved_concentration(final_radii,     coords)
    c_eff    = achieved_concentration(eff_final_radii, coords)
    print(f"Actual concentration    : {c_actual:.2%}")
    print(f"Effective concentration : {c_eff:.2%}  "
          f"(packing target - must be below ~0.87)")
    if c_eff > 0.87:
        print("  WARNING: effective concentration is very high."
              " Packing may be slow or fail to converge.")
    print()

    # ------------------------------------------------------------------
    # 3.  Progressive-growth packing on effective radii
    # ------------------------------------------------------------------
    positions, eff_radii_packed = progressive_growth_packing(
        eff_final_radii, coords, rng,
        initial_scale        = initial_radius_scale,
        initial_growth_step  = initial_growth_step,
        minimum_growth_step  = minimum_growth_step,
        growth_step_increase = growth_step_increase,
        growth_step_decrease = growth_step_decrease,
        relax_steps          = relaxation_steps_per_growth,
        final_relax_steps    = final_relaxation_steps,
        relax_factor         = relaxation_factor,
        target_max_overlap   = target_max_overlap,
        enable_relocation    = enable_relocation,
        reloc_fraction       = relocation_fraction,
        reloc_candidates     = relocation_candidates,
        n_cand_initial       = n_candidates_initial,
        save_plots           = save_growth_plots,
        plot_every           = plot_every_n_steps,
    )

    # ------------------------------------------------------------------
    # 4.  Final tiny-overlap shrink on effective radii
```

```
    # ------------------------------------------------------------------
    eff_radii_final, shrink, n_ov, max_ov, pen = final_shrink(
        positions, eff_radii_packed,
        threshold = tiny_overlap_threshold,
        safety    = final_shrink_safety,
        enable    = enable_final_shrink,
    )

    # ------------------------------------------------------------------
    # 5.  Recover actual radii  R = R_eff - min_gap/2
    # ------------------------------------------------------------------
    actual_radii = eff_radii_final - 0.5 * min_gap
    c_final      = achieved_concentration(actual_radii, coords)

    # ------------------------------------------------------------------
    # 6.  Verify minimum gap
    # ------------------------------------------------------------------
    n_viol, min_clr = check_min_gap(positions, actual_radii, min_gap)

    print()
    print("=" * 60)
    print("Final result")
    print("=" * 60)
    print(f"  Target concentration   : {ice_concentration:.2%}")
    print(f"  Final concentration    : {c_final:.2%}")
    print(f"  Number of floes        : {len(actual_radii)}")
    print(f"  Eff-radius overlaps    : {n_ov}   max = {max_ov:.3e} m")
    print(f"  Radius shrink applied  : {shrink:.3e} m")
    print()
    print(f"  Min-gap required       : {min_gap * 1000:.2f} mm")
    print(f"  Min clearance found    : {min_clr * 1000:.4f} mm")
    print(f"  Gap violations         : {n_viol}")
    if n_viol == 0:
        print("  => Min-gap condition SATISFIED for all pairs.")
    else:
        print("  => WARNING: min-gap condition violated for some pairs!")
    print("=" * 60)
    print()

    # ------------------------------------------------------------------
    # 7.  Save outputs
    # ------------------------------------------------------------------
    c_tag = f"C{int(round(c_final * 100)):02d}"

    # 7a.  Raw geometry CSV  (x, y, radius - no time column, for inspection)
    raw_path = OUTPUT_DIR /
    ↪  f"ice_floe_field_raw_{c_tag}_gap{int(min_gap*1000)}mm.csv"
    pd.DataFrame({
        "x"     : positions[:, 0],
```

```
        "y"     : positions[:, 1],
        "radius": actual_radii,
    }).to_csv(raw_path, index=False)
    print(f"Raw geometry CSV  : {raw_path.name}  ({len(actual_radii)} floes)")

    # 7b.  Snapshot injection CSV  (all floes at t = time_first_impact)
    #      Use this in STAR-CCM+ to verify the field visually before
    #      switching to the time-bunched per-velocity files.
    placed_floes = [
        {"x": float(positions[i, 0]),
         "y": float(positions[i, 1]),
         "radius": float(actual_radii[i])}
        for i in range(len(actual_radii))
    ]
    save_snapshot_injection_csv(
        placed_floes      = placed_floes,
        ice_concentration = ice_concentration,
        min_gap           = min_gap,
        thickness         = thickness,
        time_first_impact = time_first_impact,
        output_dir        = OUTPUT_DIR,
    )

    # ------------------------------------------------------------------
    # 8.  Plots
    # ------------------------------------------------------------------
    _plot_field(
        positions, actual_radii, coords,
        title    = (f"Ice floe field - C = {c_final:.1%} | "
                    f"{len(actual_radii)} floes | gap >= {min_gap*1000:.0f}
                    ↪  mm"),
        filepath = OUTPUT_DIR /
        ↪  f"ice_floe_field_{c_tag}_gap{int(min_gap*1000)}mm.png",
        show     = False,
    )

    _plot_histogram(
        actual_radii, mu_diameter, sigma_diameter,
        min_diameter, max_diameter,
        filepath = OUTPUT_DIR / f"diameter_histogram_{c_tag}.png",
    )

    # ------------------------------------------------------------------
    # 9.  Export one time-bunched injection CSV per velocity
    # ------------------------------------------------------------------
    print("\nExporting injection CSVs")
    print("------------------------")
    all_files = []
    for v in velocities:
```

```
        df_inj, fp = save_injection_csv(
            velocity          = v,
            placed_floes      = placed_floes,
            coords            = coords,
            ice_concentration = ice_concentration,
            min_gap           = min_gap,
            thickness         = thickness,
            duration_impact   = duration_impact,
            time_first_impact = time_first_impact,
            time_step         = time_step,
            x_injection       = x_injection,
            Lpp               = Lpp,
            g                 = g,
            output_dir        = OUTPUT_DIR,
        )
        all_files.append({
            "velocity_mps"    : v,
            "n_injected_rows" : len(df_inj),
            "filepath"        : fp.name,
        })

    summary_path = OUTPUT_DIR / "injection_files_summary.csv"
    pd.DataFrame(all_files).to_csv(summary_path, index=False)

    print()
    print("Done. Summary:")
    print(pd.DataFrame(all_files).to_string(index=False))
    print(f"\nSummary CSV: {summary_path}")

    elapsed = time.perf_counter() - t_start
    print(f"\nTotal run time : {int(elapsed // 60)} min {elapsed % 60:.1f} s")

    plt.show()
```